

Calcium imaging figure guide

Stage 1: quality control and motion

A plot-by-plot walkthrough of every figure in plots/. For each one: what the analysis is, how it was computed, and how to read it. Open this file in a Markdown viewer from inside analysis_calcium/ so the inline images resolve.

The recording in one paragraph

Pan-neuronal GCaMP6m (elav promoter) in a juvenile *Helobdella* leech ganglion, imaged at 10x on a confocal. 692 frames, 700x700 px, single channel, frame rate $f_s = 2.882$ Hz (0.347 s/frame), total ~240 s. Annotated "10x juvenile after dopamine pharyngals", movement "no". Because it is low magnification and a preliminary line, everything is interpreted at the ganglion / regional level, not as single neurons. The dominant phenomenon is a single, strong, clock-like, ganglion-wide calcium oscillation at 0.372 Hz (period 2.69 s) that is sustained for the whole recording and synchronous across the ganglion.

How the numbers are produced (shared definitions)

- dF/F : for each region/patch, F_0 is a rolling 10th-percentile baseline (~60 s window); $dF/F = (F - F_0) / F_0$. Neuropil was subtracted ($F - 0.7 \cdot F_{neu}$) before this.
- Ganglion signal: mean raw F inside a tissue mask (the bright ganglion band), converted to dF/F . This is the primary high-level trace (derived/ganglion_signal.npy).
- Patches: 180 ROIs from a correlation-image blob+watershed segmentation (the primary set). suite2p's 24 ROIs are kept as a cross-check. At 10x these are "active patches", not single cells.
- Dominant frequency: peak of the Welch power spectrum of the ganglion signal.
- Cycle peaks: scipy find_peaks on the ganglion signal; 88 cycles detected.

What we did and how to interpret

What we did: the analysis pipeline

The recording was analysed in five stages, each a script in scripts/, followed by an intricate visualization pass. All outputs are derived from the single raw movie; no manual edits to traces were made.

- Stage 1 (QC and motion): measured photobleaching, rigid drift (phase cross-correlation of every frame to a reference), non-rigid block-wise jitter, and built the local correlation image that reveals cell bodies.
- Stage 2 (suite2p): ran suite2p with registration on to detect ROIs and extract fluorescence. It accepted only 24 ROIs, far fewer than the visible somata, so it was kept only as a cross-check.
- Stage 2b (correlation-blob segmentation): the primary method. Detected somata on the correlation image times the std projection (peak detection plus watershed to split touching cells), giving 180 active patches, and extracted dF/F with local-annulus neuropil subtraction.
- Stage 3 (temporal): characterised the rhythm of the whole-ganglion mean signal with Welch spectra, a spectrogram, autocorrelation, peak detection and a Hilbert transform.
- Stage 4 (spatial): mapped per-pixel oscillation power and phase, tested for a traveling wave, and measured region-by-region coherence.
- Intricate pass: 25 detailed multi-panel figures (fig_*.py) plus a cycle-averaged movie, covering population, spectral, spatial, cycle-resolved and dashboard views.

Units note: this is 10x, a preliminary line, so all signals are read at the ganglion / regional level, not as single neurons. The 180 patches are active regions, not confirmed cells.

Motion correction: not needed (verified)

Whether to motion-correct was decided from measurements, not assumed.

- Rigid drift was 0.00 px across all 692 frames (cross-checked at full resolution). The preparation is stationary; the movement flag no is correct.
- Non-rigid jitter was 0.00 px in a 3x3 block check. No local warping.
- Photobleaching was 0.5 to 0.8% over 240 s, negligible; dF/F absorbs it. No saturation, no dropped frames.

Conclusion: analysis runs directly on the raw movie with no interpolation artifacts. The moderate frame-to-frame correlation (~ 0.55) is real calcium dynamics, not motion, and must not be mistaken for a motion problem. The dataset recordings flagged leech moves or david moves are the ones that would need correction, ideally using their palmmcherry static channel.

Headline result and how to interpret it

The ganglion shows one strong, clock-like, ganglion-wide calcium oscillation at 0.37 Hz (period 2.7 s) that is sustained undamped for the full ~240 s and is spatially synchronous across the whole ganglion.

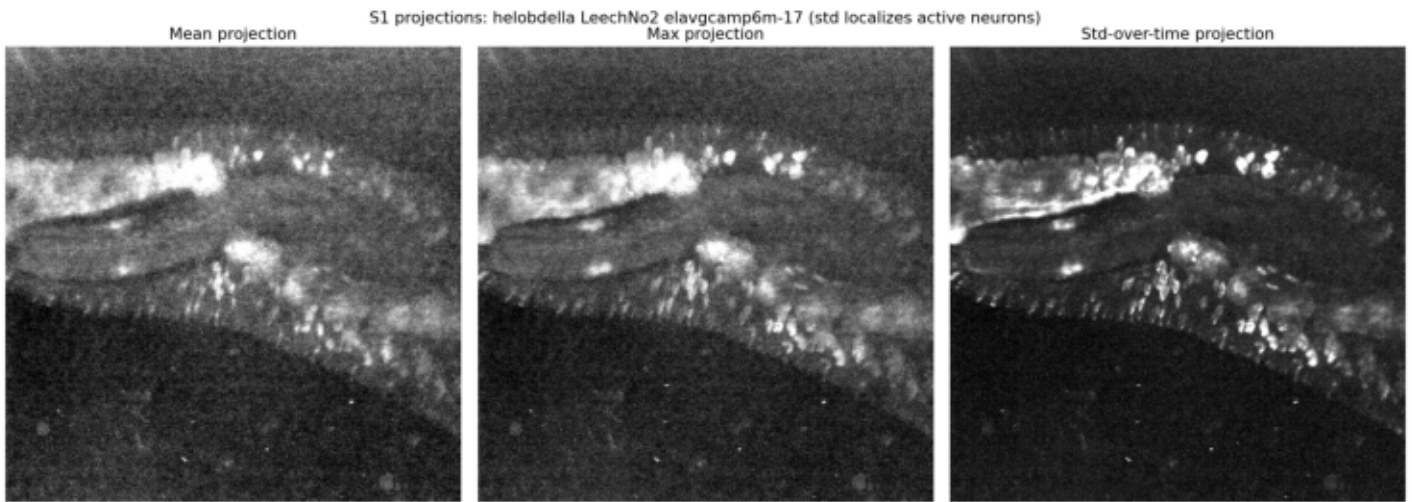
- Frequency: dominant peak 0.372 Hz with a harmonic at ~0.73 Hz (a real periodic, non-sinusoidal waveform), ~230x above the spectral floor, well below the 1.44 Hz Nyquist so the fundamental is not aliased.
- Regularity: 88 cycles, inter-peak interval CV 0.077, a tight Poincare cluster, and a stable spectrogram band. A very clock-like rhythm.
- Synchrony: vertical kymograph stripes, whole-ganglion brightening and dimming together in the phase montage, and a flat phase-vs-position slope. Inter-region coherence at 0.37 Hz is ~0.88. No traveling wave.
- Nuance: the rhythm is carried by a strong subset. About 93% of patches peak within 0.05 Hz of 0.37 Hz and ~40% follow the ganglion mean at $r > 0.5$; the rest are noisy (low-photon 10x). Pooling the rhythmic subset is what yields the clean ganglion signal, so all-patch pairwise correlation looks low while region-level coherence is high.

How to interpret: a single synchronous ganglion-wide oscillation points to a shared network drive rather than independent local pacemakers. The file is labelled after dopamine pharyngals, but there is no pre-dopamine baseline here, so no causal claim about dopamine can be made. The 0.37 Hz rhythm is a candidate motor or visceral rhythm; identifying it needs a labelled baseline and a behavioural or electrophysiological readout.

Figure walkthrough

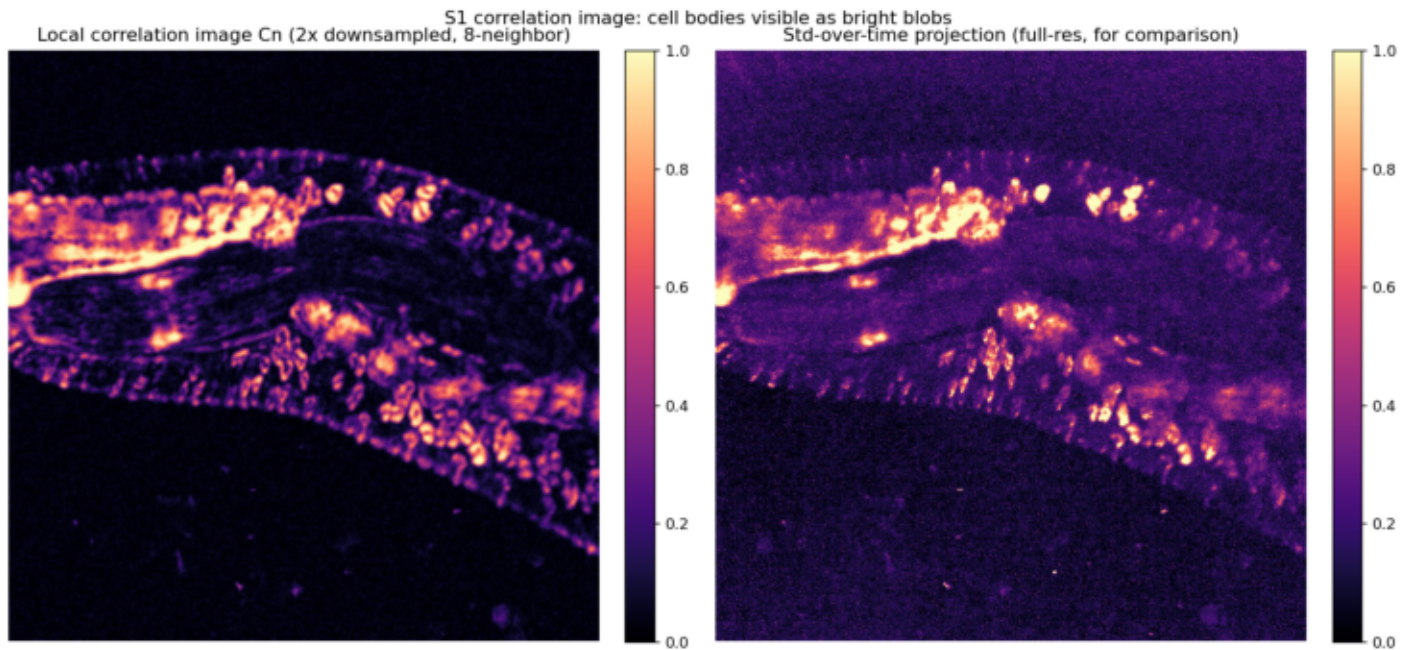
every plot, with how it was made and how to read it

s1_projections.png



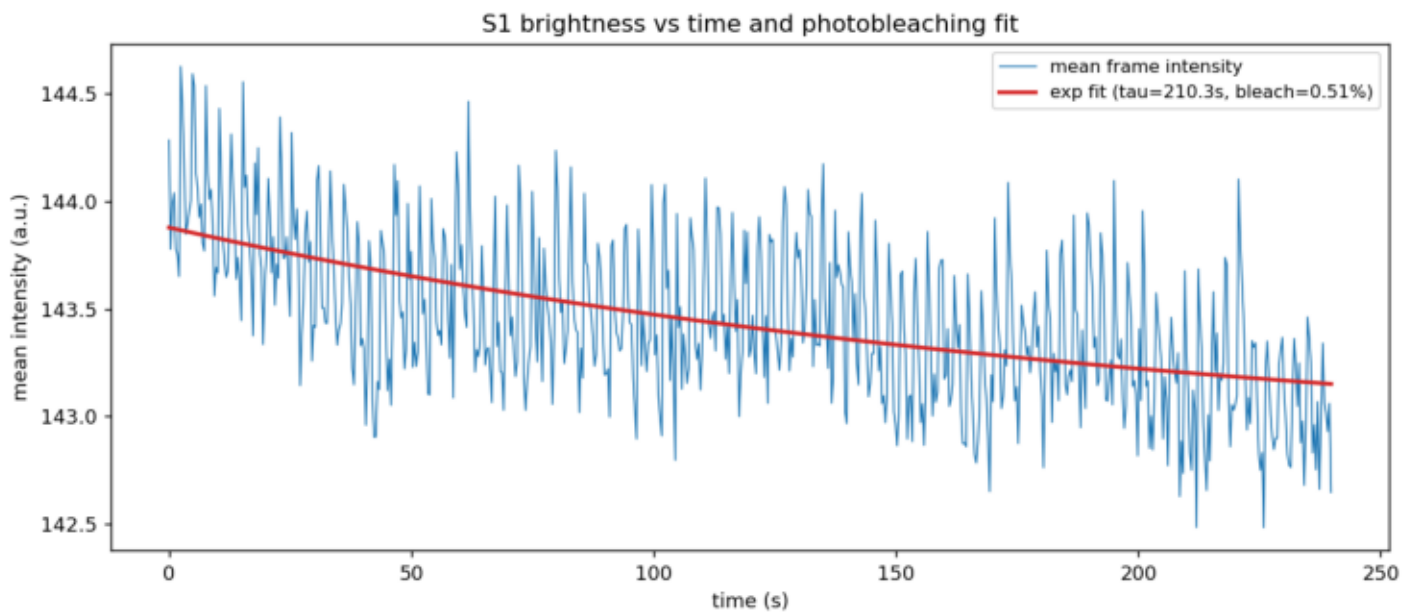
- What: three summary images of the movie collapsed over time: the mean, the maximum, and the standard deviation at each pixel.
- How: per-pixel mean/max/std across all 692 frames, with percentile contrast.
- How to read: the mean shows anatomy (where tissue is). The std image is the most useful: bright pixels are where intensity **changes** over time, i.e. active neurons. Here the std and max highlight a ganglionic band in the upper-central field.

s1_correlation_image.png



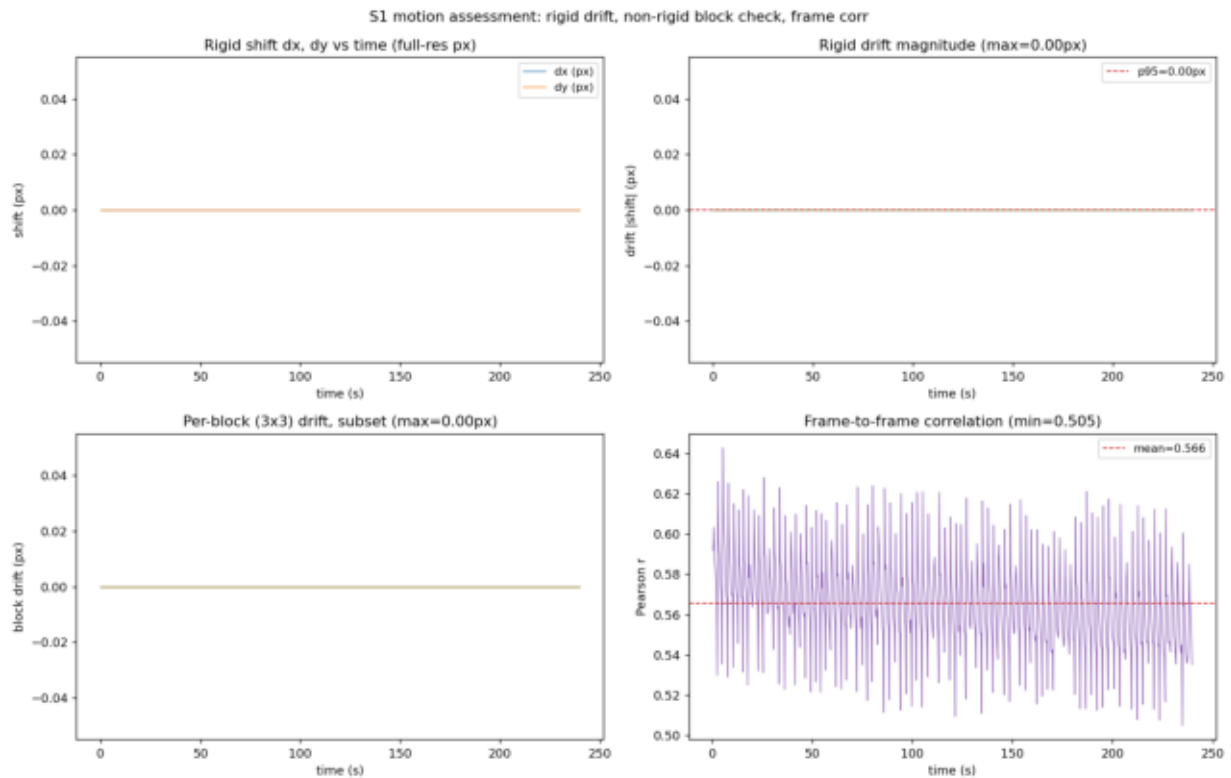
- What: the local correlation image (Cn), the cleanest map of cell bodies, beside
- the std projection.
- How: each pixel's average temporal correlation with its 8 neighbors. Cells, whose
- pixels rise and fall together, light up; uncorrelated background stays dark.
- How to read: bright blobs = candidate somata. This image seeds segmentation. The
- bright band is the ganglion; the dim periphery is background.

s1_brightness_bleach.png



- What: photobleaching check.
- How: mean frame intensity vs time (all 692 frames) with an exponential-decay fit.
- How to read: a steep downward curve would mean bleaching that needs detrending.
- Here the line is nearly flat (0.5 to 0.8% loss over 240 s), so no detrending is needed; dF/F absorbs it.

s1_motion.png

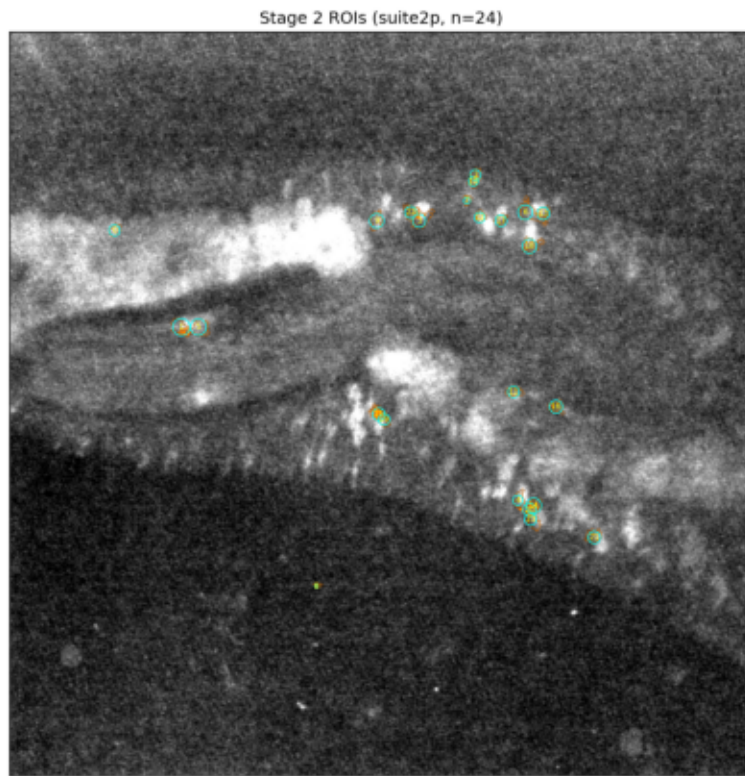


- What: the motion-correction decision evidence.
- How: rigid frame-to-reference drift via phase cross-correlation for every frame
- (dx, dy, magnitude); a 3x3 block-wise drift check for non-rigid jitter; and
- frame-to-frame correlation over time.
- How to read: drift traces flat at 0 px = no translation; flat block drift = no
- local warping. The conclusion is no motion correction needed. The ~0.55
- frame-to-frame correlation reflects real calcium dynamics, not motion (do not mistake
- it for a motion problem).
- ---

Stage 2 and 2b: segmentation and signal extraction

s2_rois_on_mean.png

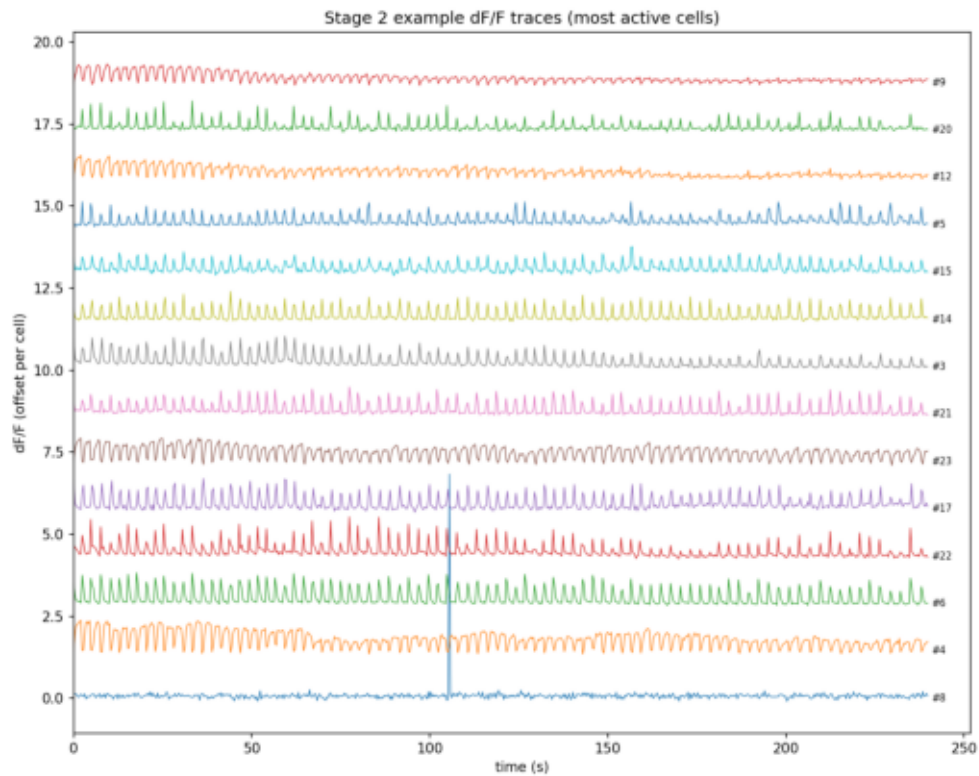
Stage 2 and 2b: segmentation and signal extraction



- What: suite2p's detected ROIs ($n = 24$) on the mean image.
- How: suite2p run with registration on, tau 1.0, diameter 15, relaxed threshold.
- How to read: each circle is an accepted ROI. Note how few there are relative
- to the visible somata, which motivated the alternative method below.

s2_example_traces.png

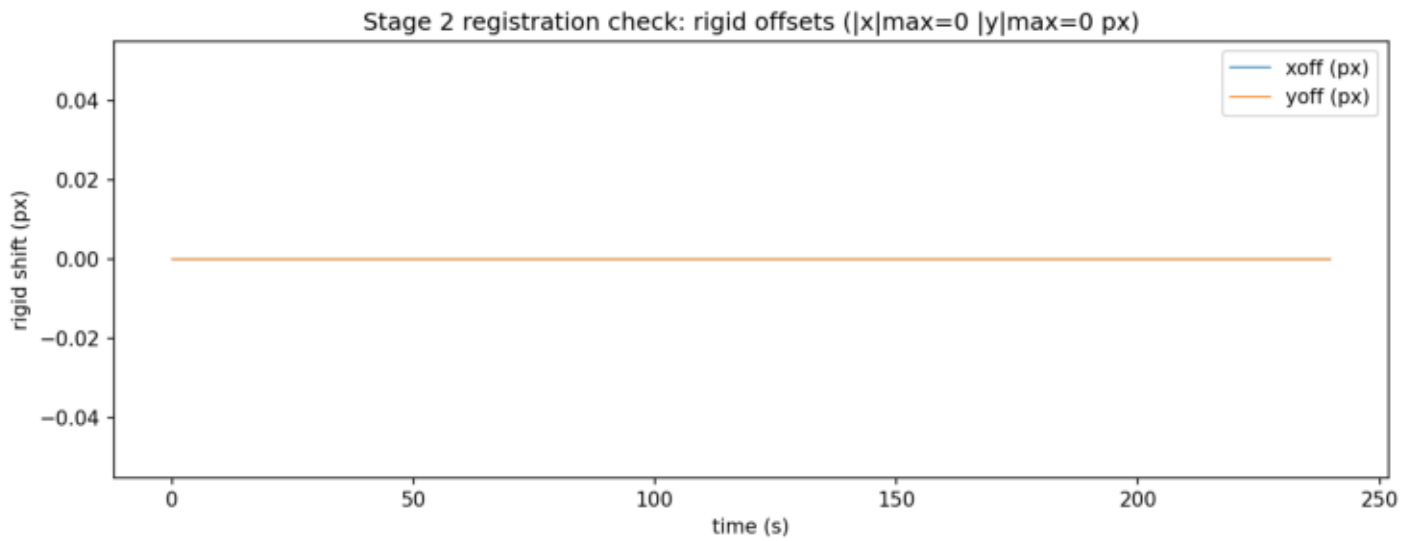
Stage 2 and 2b: segmentation and signal extraction



- What: dF/F traces of the most active suite2p ROIs, stacked with vertical offsets.
- How to read: x is seconds; each row is one ROI. The regular up-down ripples are
- the calcium rhythm; nearly every ROI shows it.

s2_registration_check.png

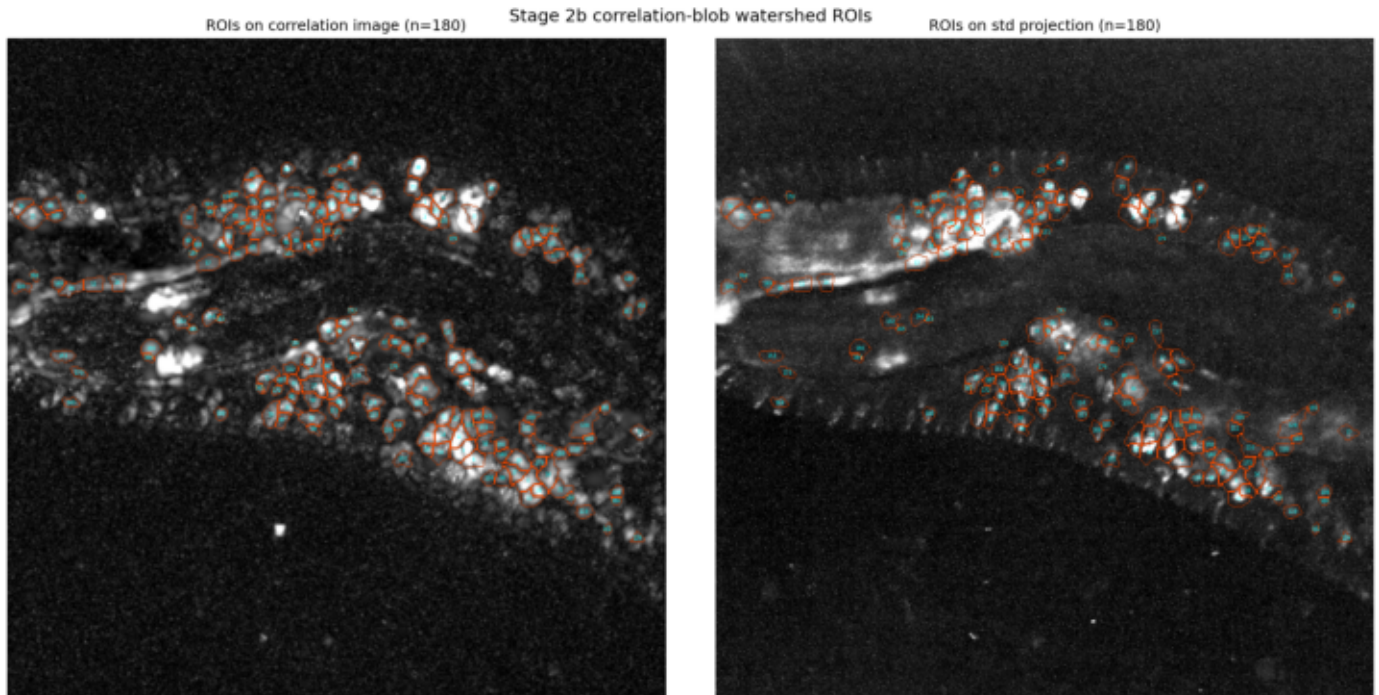
Stage 2 and 2b: segmentation and signal extraction



- What: confirmation that registration was a no-op.
- How: suite2p rigid x/y offsets per frame.
- How to read: both traces flat at 0 = the algorithm found nothing to correct,
- independently confirming Stage 1's "no motion".

s2b_rois_on_corr.png

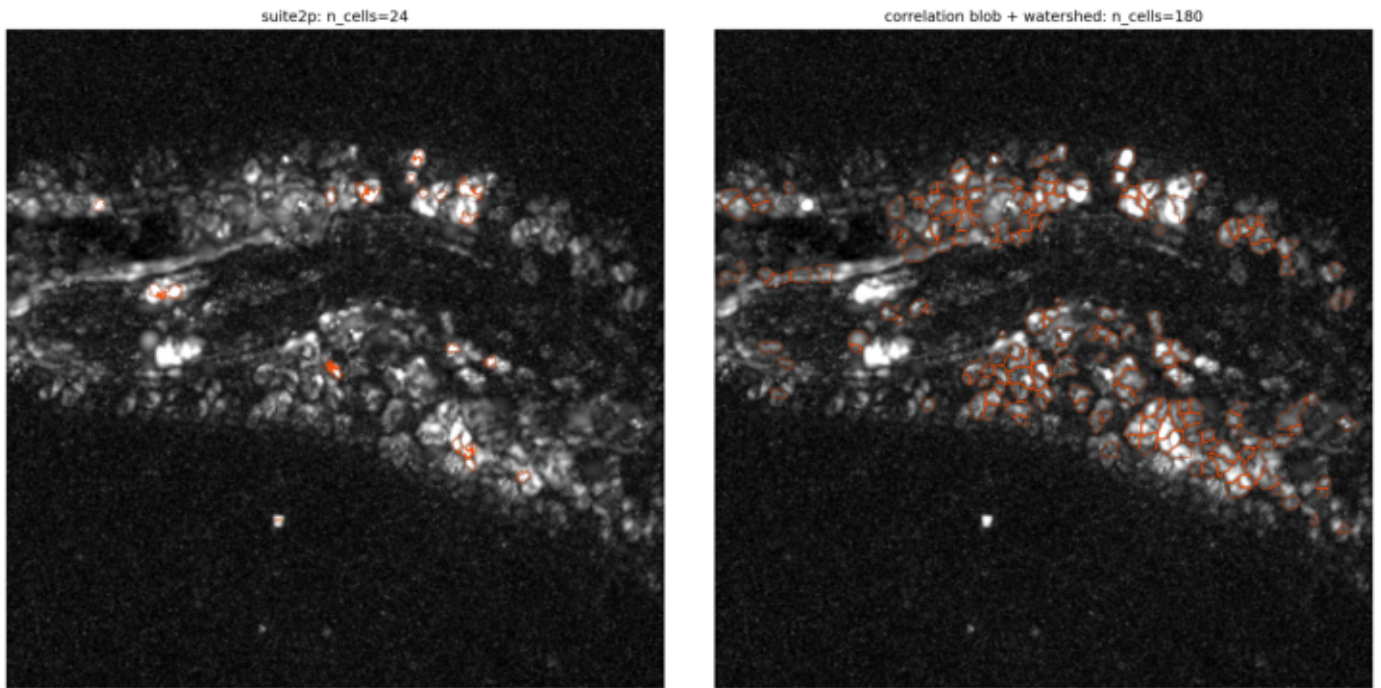
Stage 2 and 2b: segmentation and signal extraction



- What: the primary segmentation, 180 ROIs from correlation-blob + watershed.
- How: detection image = normalized correlation image x std projection; tissue mask
- by Otsu; soma seeds by peak_local_max (min distance ~ soma radius); watershed to split
- touching somata; filtered by area and brightness.
- How to read: numbered outlines tile the active ganglion band. This is 7.5x more
- ROIs than suite2p and is the set used for all downstream patch analyses.

s2b_method_comparison.png

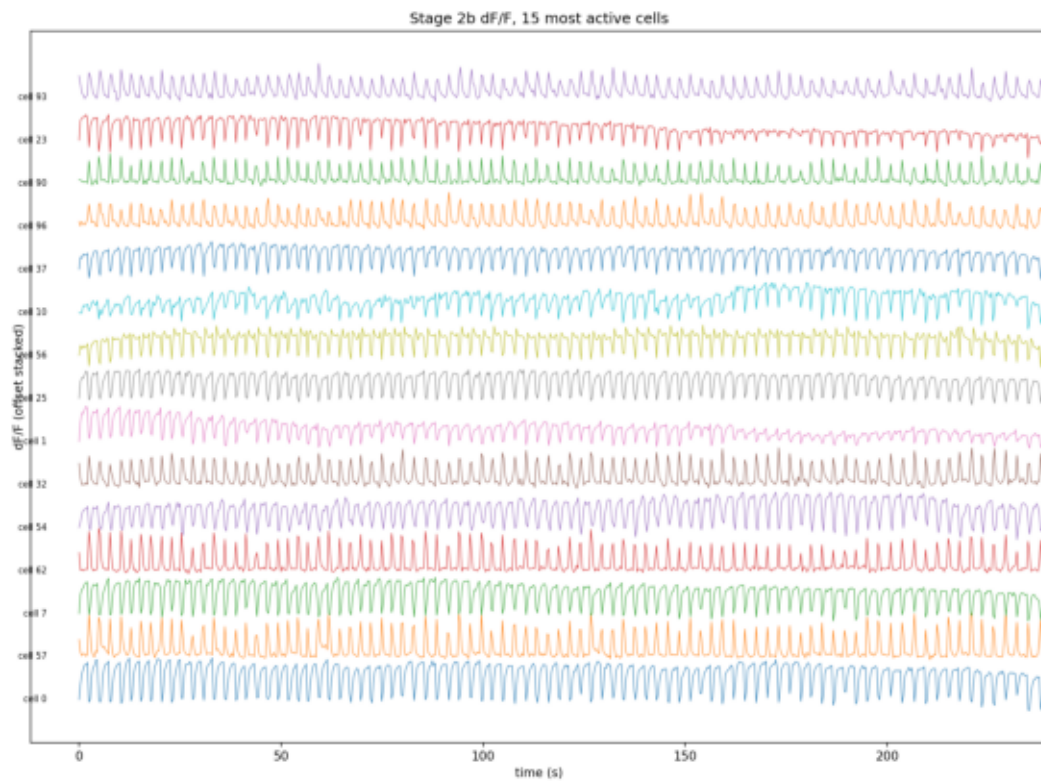
Stage 2 and 2b: segmentation and signal extraction



- What: suite2p (24) vs correlation-blob (180) side by side on the same image.
- How to read: the under-segmentation by suite2p is visually obvious; the blob method
- covers cells suite2p left empty. Both still recover the same rhythm, so the result does
- not depend on the choice.

s2b_example_traces.png

Stage 2 and 2b: segmentation and signal extraction

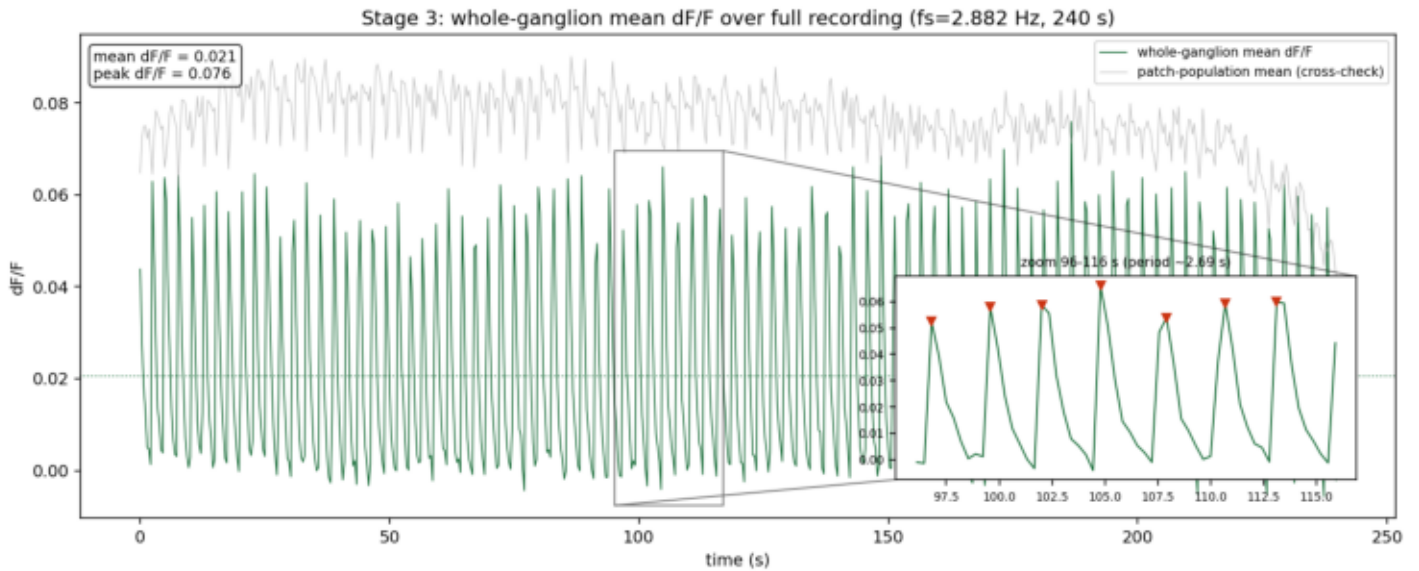


- What: dF/F of the 15 most active patches from the primary method.
- How to read: stacked traces in seconds; the shared, near-identical ripple across
- rows is the ganglion-wide rhythm.
- ---

Stage 3: temporal structure of the rhythm

s3_ganglion_signal.png

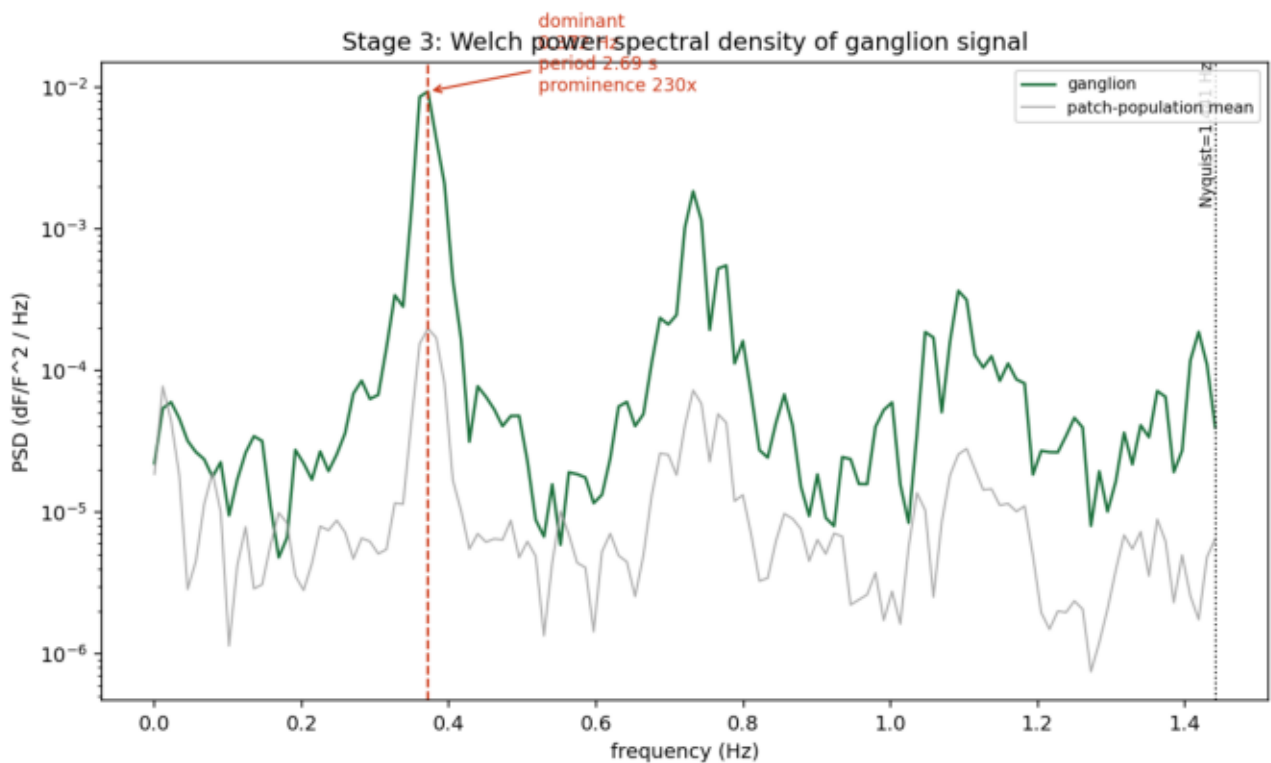
Stage 3: temporal structure of the rhythm



- What: the whole-ganglion mean dF/F over the full recording, with a 20 s zoom inset.
- How to read: the sustained regular oscillation runs the entire 240 s without
- damping; the inset shows the waveform shape of single cycles. Peak $dF/F \sim 0.08$ (modest,
- expected for 10x bulk imaging).

s3_spectrum.png

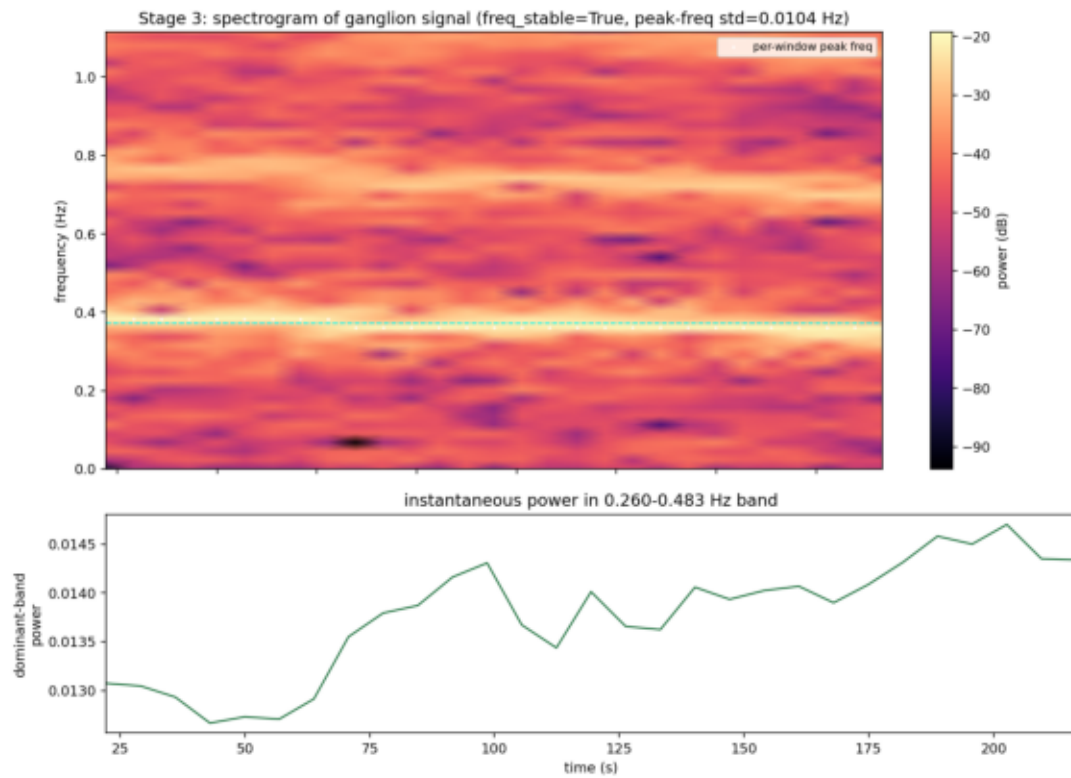
Stage 3: temporal structure of the rhythm



- What: Welch power spectral density of the ganglion signal (and patch-population mean).
- How to read: the sharp peak at 0.372 Hz is the rhythm; the smaller bump at ~0.73 Hz is its harmonic (proof the waveform is periodic but not a pure sinusoid). The peak stands ~230x above the broadband floor. The dotted Nyquist line (1.44 Hz) is far to the right, so the fundamental is not aliased.

s3_spectrogram.png

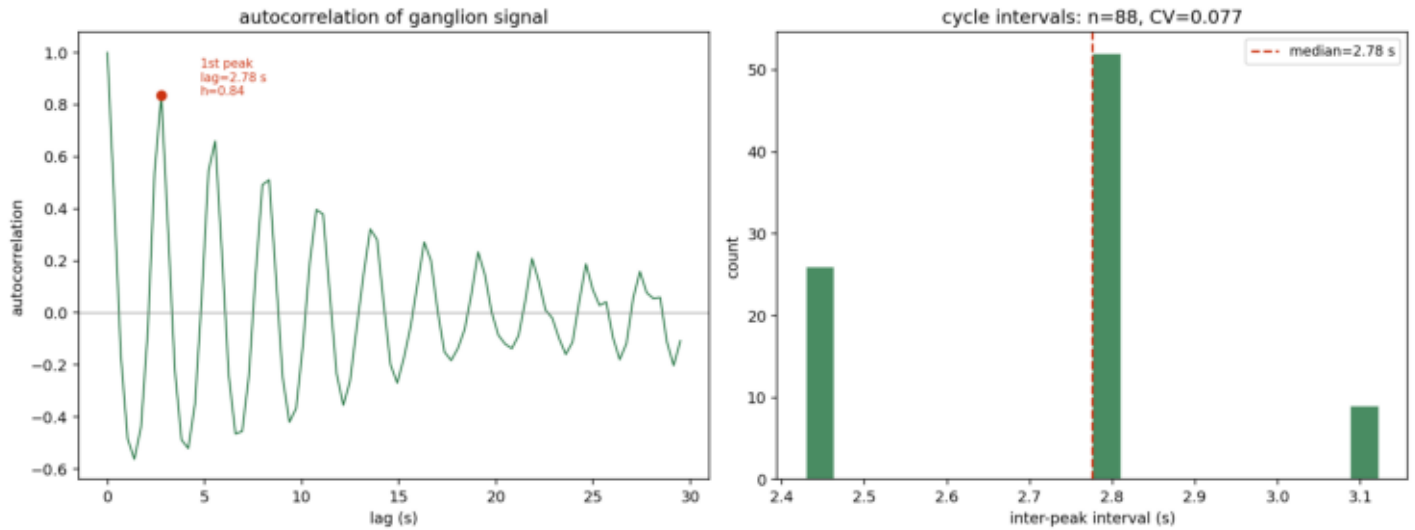
Stage 3: temporal structure of the rhythm



- What: how the frequency content evolves over time.
- How: sliding-window FFT (time on x, frequency on y, power as color).
- How to read: a single horizontal band fixed at ~ 0.37 Hz across the whole recording
- means the rhythm is frequency-stable, not drifting.

s3_autocorr_regularity.png

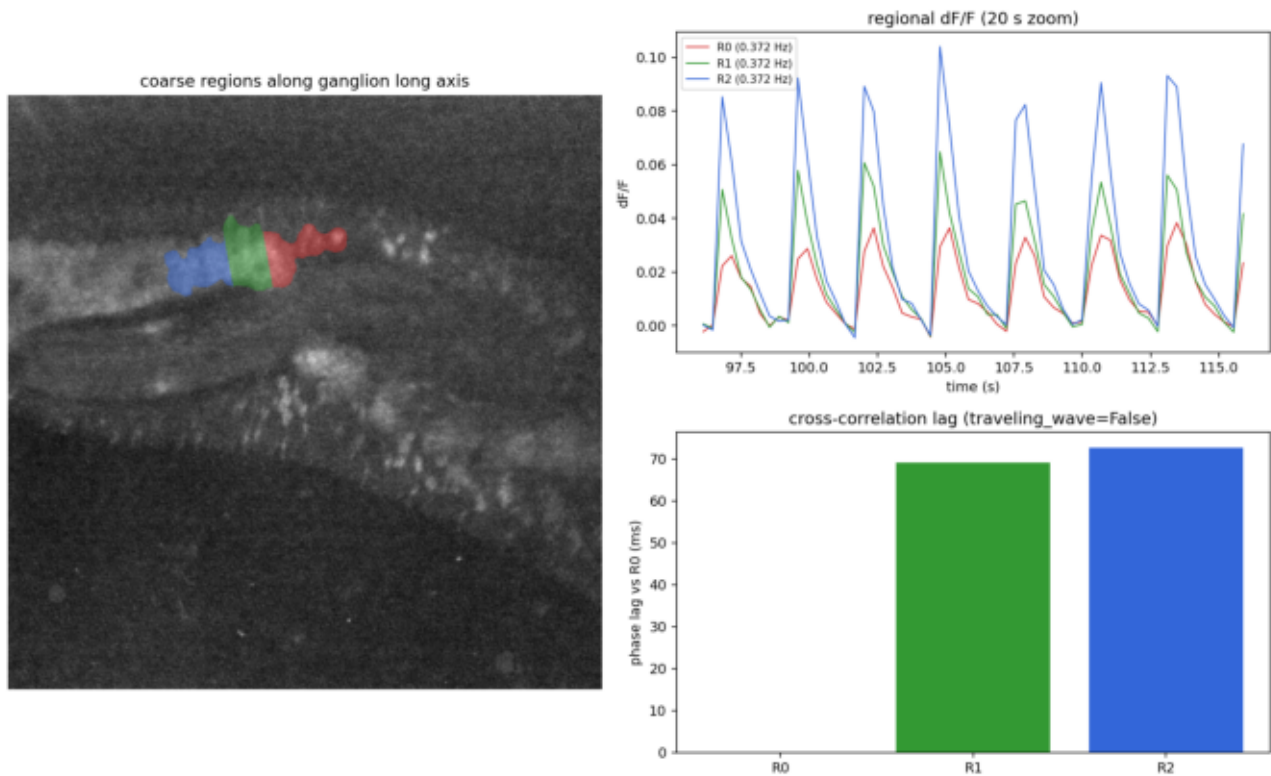
Stage 3: temporal structure of the rhythm



- What: how clock-like the rhythm is.
- How: autocorrelation of the ganglion signal plus the distribution of inter-peak intervals.
- How to read: tall, evenly spaced autocorrelation side-peaks = strong periodicity;
- the first side-peak at ~ 2.78 s is the period. The narrow interval histogram (CV 0.077)
- confirms very low cycle-to-cycle jitter.

s3_regional_comparison.png

Stage 3: temporal structure of the rhythm

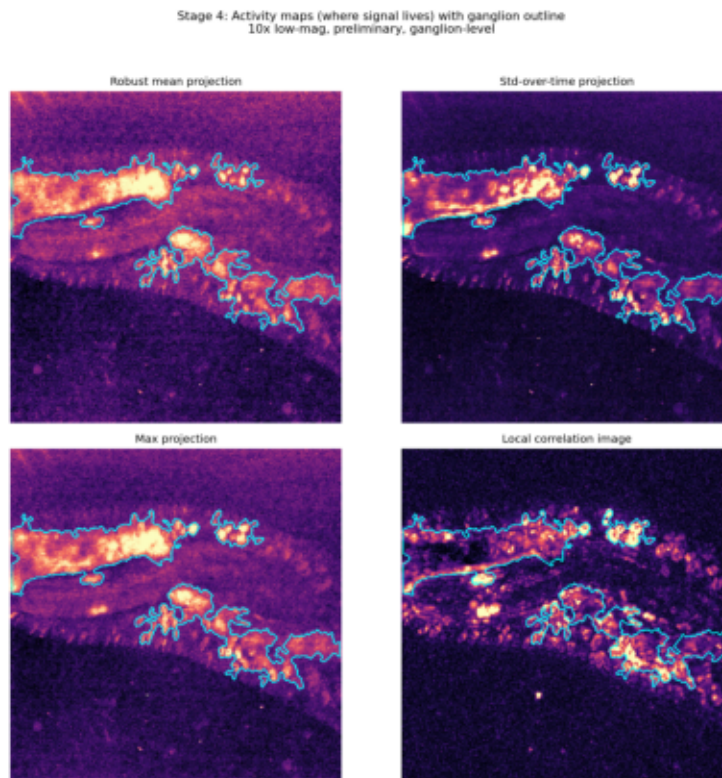


- What: do coarse parts of the ganglion share the rhythm and its timing.
- How: split the band into a few regions (anterior/mid/posterior), compare their
- traces, frequencies, and cross-correlation phase lags.
- How to read: identical frequencies and sub-frame (~ 70 ms) lags = the regions are
- synchronous, with no traveling-wave gradient.
- ---

Stage 4: spatial structure of the rhythm

s4_activity_maps.png

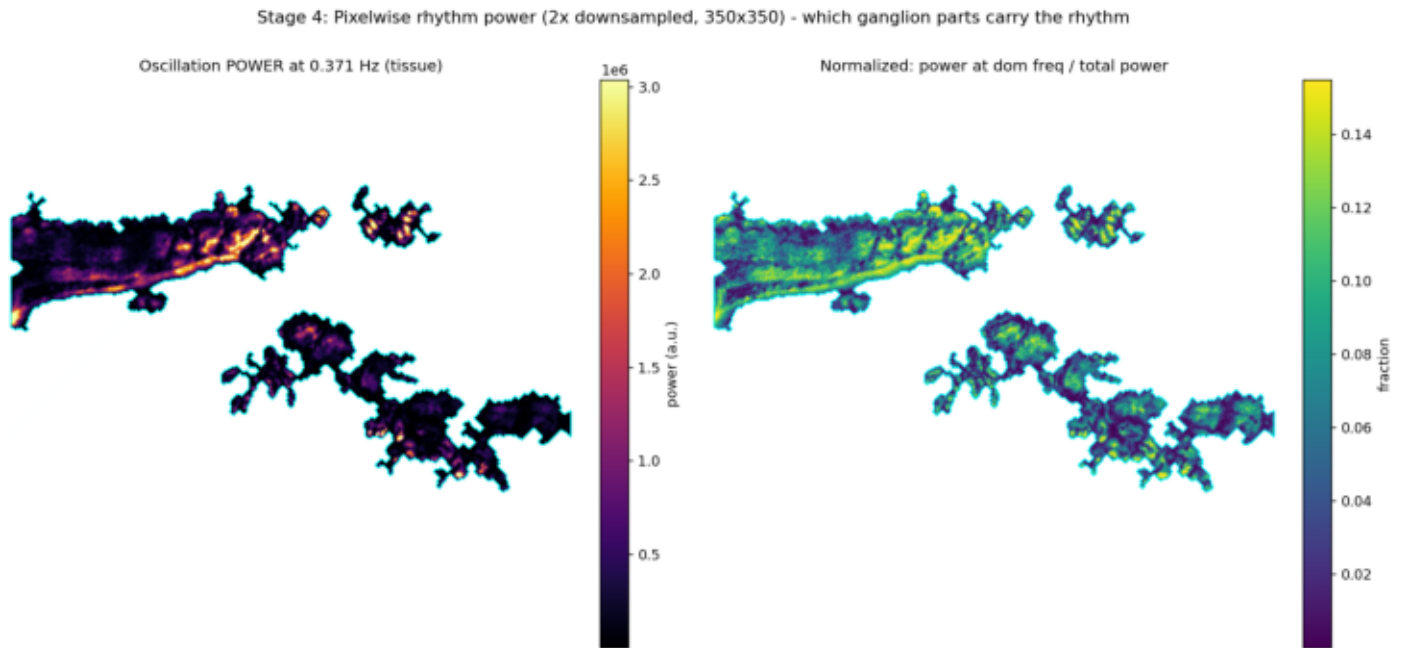
Stage 4: spatial structure of the rhythm



- What: where signal lives, as a panel of mean / std / max / correlation projections
- with the ganglion outline.
- How to read: signal is confined to the elongated ganglion band (~13% of the field);
- the periphery is background.

s4_pixelwise_rhythm_power.png

Stage 4: spatial structure of the rhythm

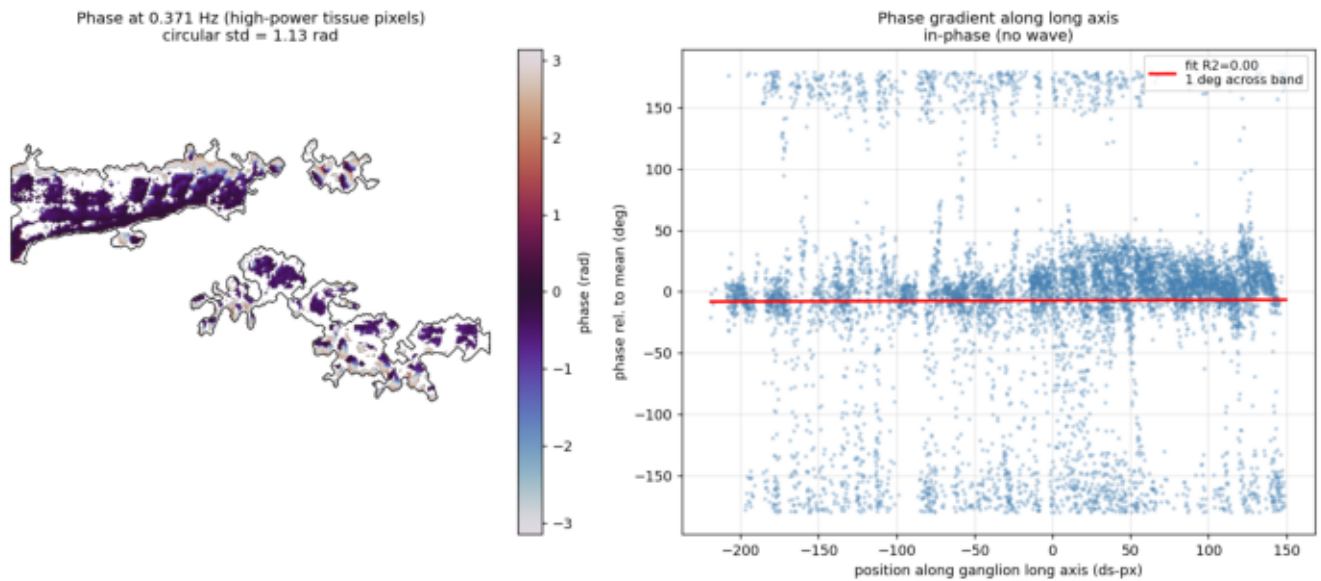


- What: which pixels carry the 0.37 Hz rhythm.
- How: per-pixel FFT power at the dominant frequency (left absolute, right normalized by total power).
- How to read: bright = oscillating at 0.37 Hz. The rhythm is spread across the whole band (~93% of tissue pixels), not localized to one spot.

s4_phase_map.png

Stage 4: spatial structure of the rhythm

Stage 4: Phase structure of the ganglion oscillation

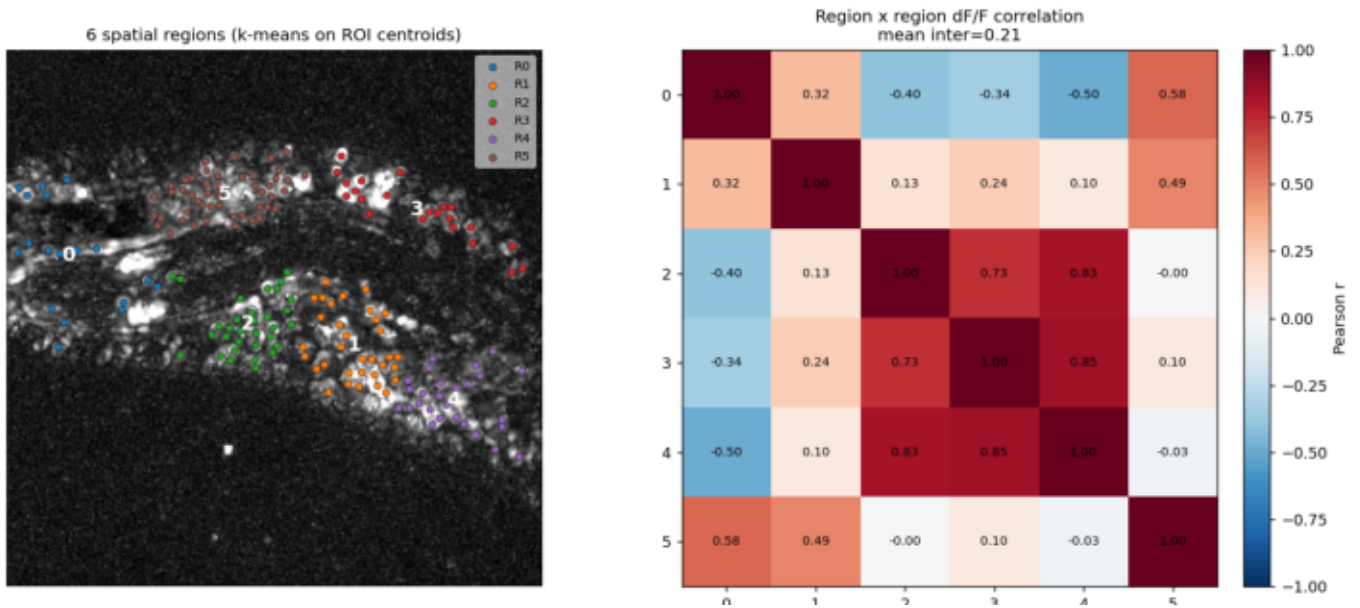


- What: the timing (phase) of the oscillation across space.
- How: angle of each pixel's FFT coefficient at 0.37 Hz, cyclic colormap, masked to high-power tissue.
- How to read: roughly uniform color = the ganglion oscillates in phase. A smooth color gradient would indicate a traveling wave; here there is none.

s4_regional_correlation.png

Stage 4: spatial structure of the rhythm

Stage 4: Regional coherence - one unit vs distinct territories

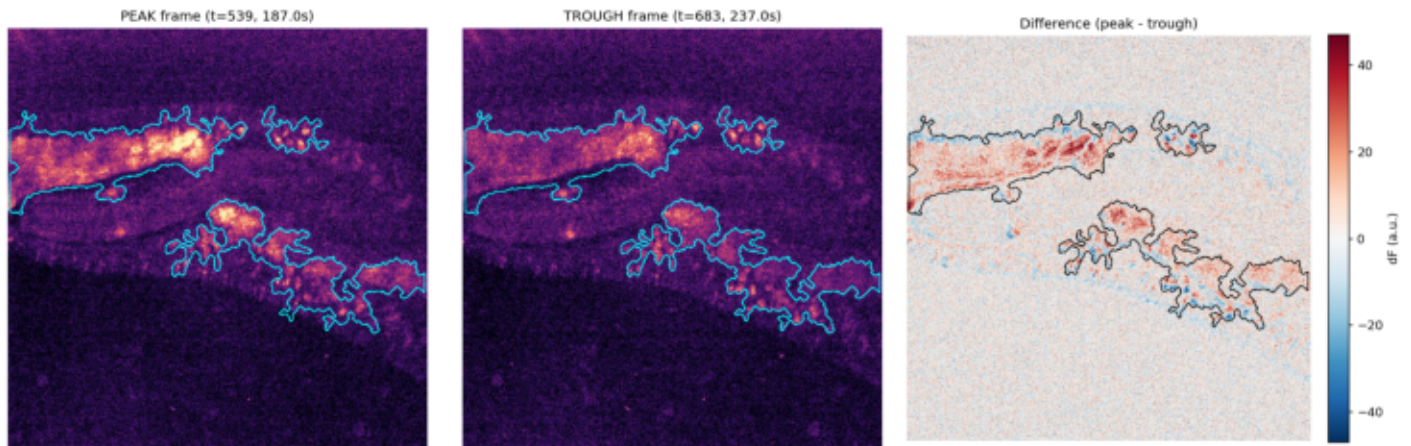


- What: are there distinct functional territories.
- How: tile the tissue into ~6 regions, correlate their mean traces.
- How to read: mostly positive correlations = one coherent rhythm; the spread (some weak/negative pairs) reflects varying signal-to-noise, not clean separate modules.

s4_peak_trough_frames.png

Stage 4: spatial structure of the rhythm

Stage 4: Peak vs trough raw frames - rhythm is real fluorescence modulation (motion already ruled out)

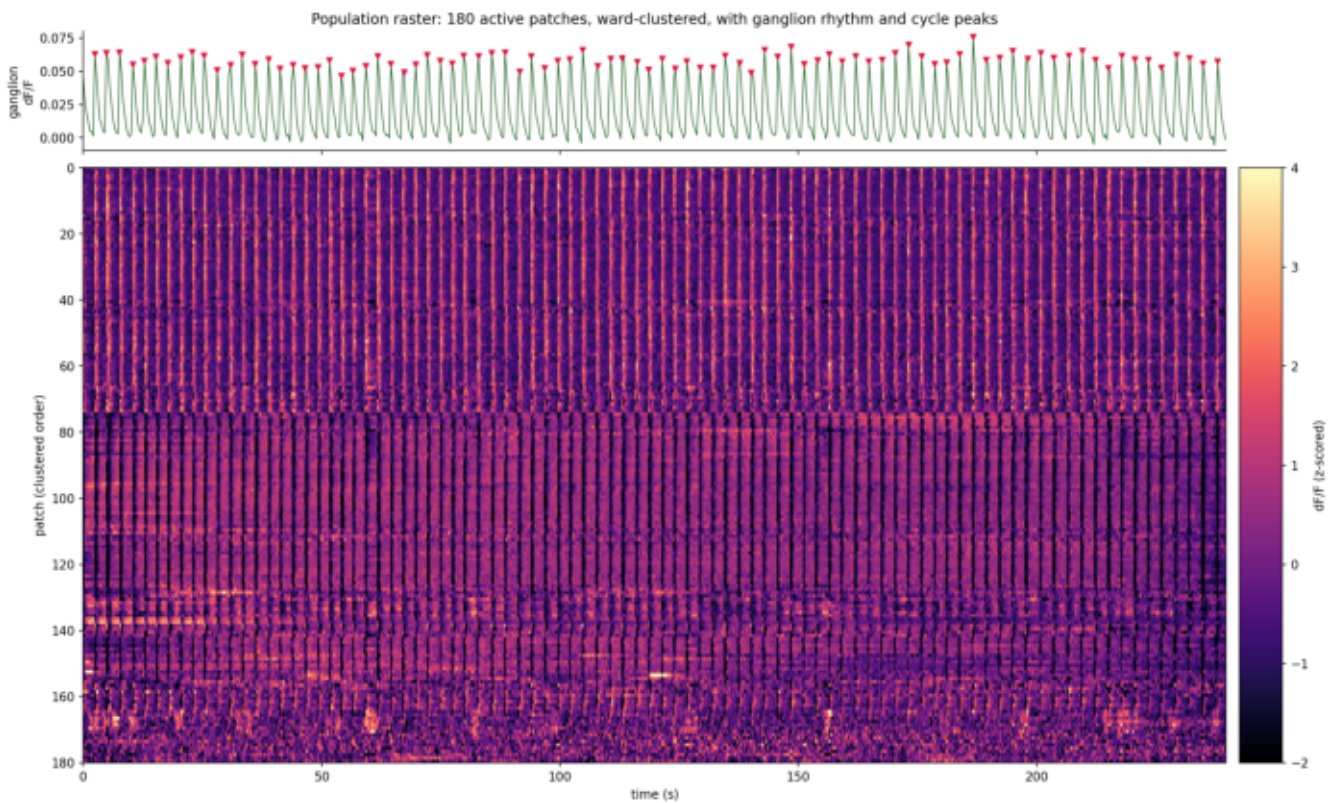


- What: a sanity check that the rhythm is real fluorescence, not noise.
- How: a raw frame at an oscillation peak, one at a trough, and their difference.
- How to read: the ganglion is uniformly brighter at the peak; the positive
- difference image confirms a genuine brightness modulation (motion already excluded).
- ---

Intricate set: population / temporal (fig_pop_*)

fig_pop_raster.png

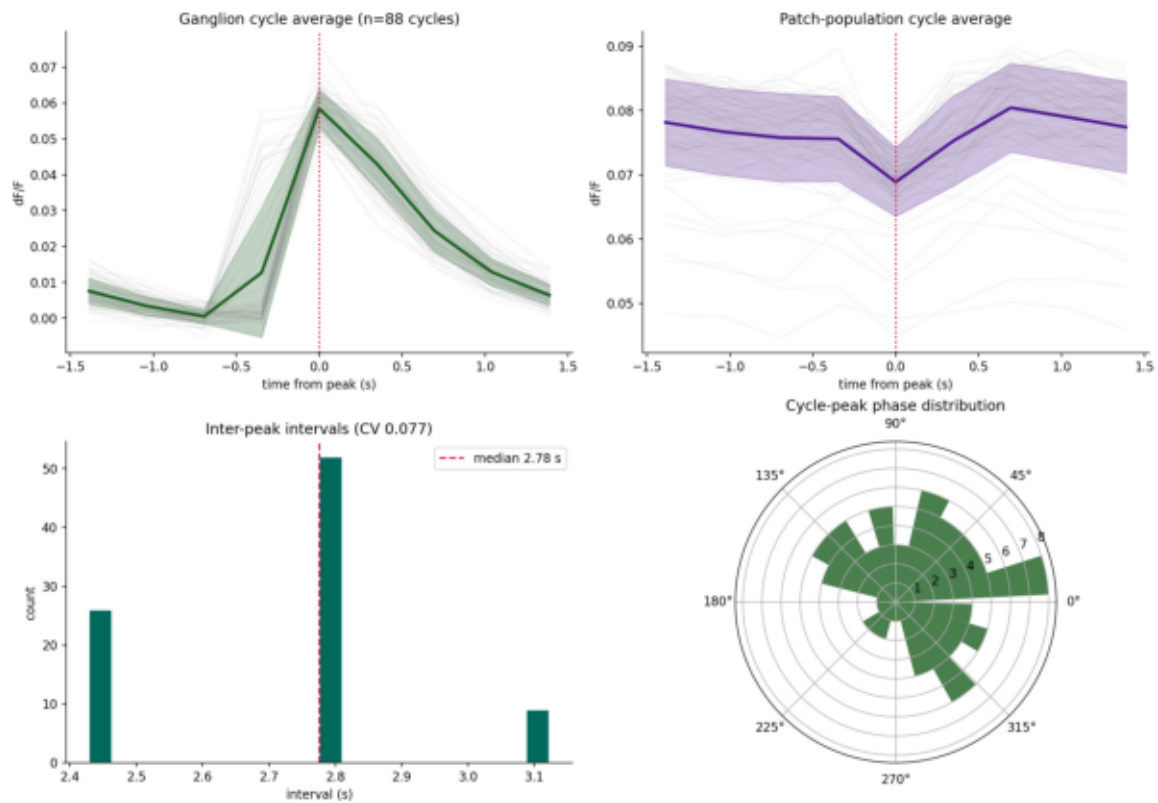
Intricate set: population / temporal (fig_pop_*)



- What: the whole population at a glance.
- How: heatmap of all 180 patch dF/F (z-scored per patch), patches ordered by ward
- hierarchical clustering on correlation distance; the ganglion mean with cycle peaks
- runs along the top.
- How to read: each row is a patch, x is time. Vertical banding that lines up across
- rows = synchronized activity. The clustering groups patches with similar time courses.

fig_pop_cycle_average.png

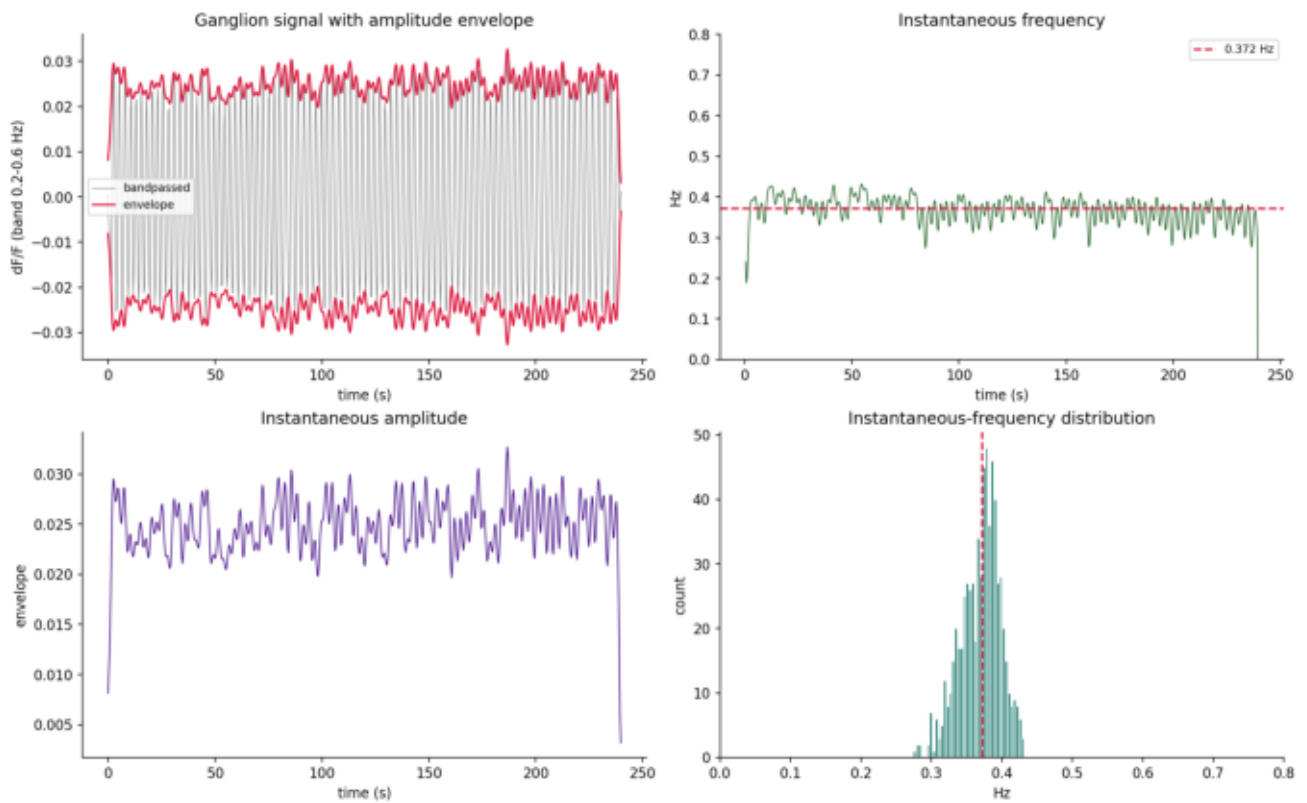
Intricate set: population / temporal (fig_pop_*)



- What: the average shape of one oscillation cycle.
- How: cut the signal into windows centered on each detected peak, overlay all 88
- cycles (grey) and their mean +/- std (bold); plus the peak-phase histogram and a polar
- plot.
- How to read: tight overlap of the grey traces = highly reproducible waveform. The
- mean cycle shows the rise/fall asymmetry. The polar plot shows peak timing concentration.

fig_pop_instantaneous.png

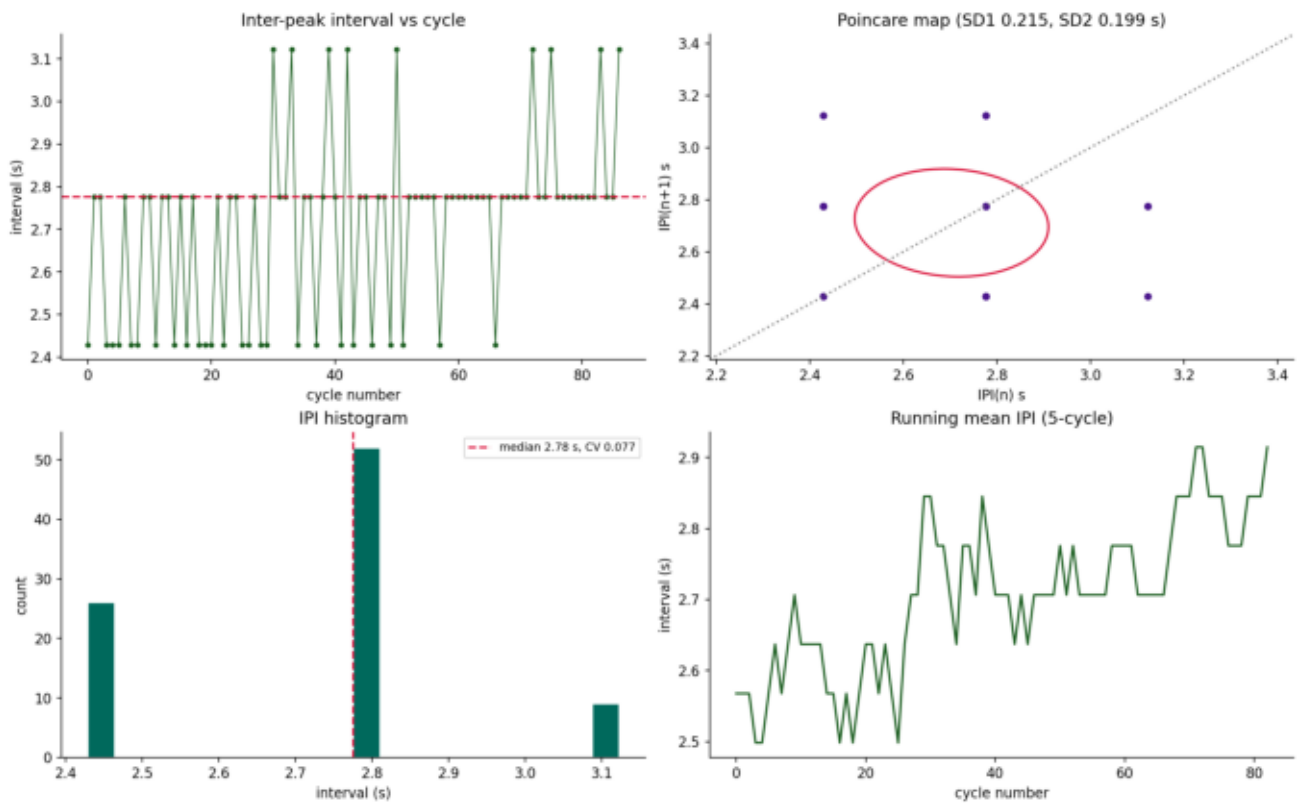
Intricate set: population / temporal (fig_pop_*)



- What: instantaneous amplitude and frequency via the Hilbert transform.
- How: bandpass the ganglion signal 0.2 to 0.6 Hz, take the analytic signal; envelope
- = amplitude, derivative of phase = instantaneous frequency.
- How to read: the instantaneous frequency hugging 0.37 Hz (panel b and its
- histogram) is strong evidence of a stable single oscillator; the envelope (panels a, c)
- shows slow amplitude waxing and waning.

fig_pop_interval_dynamics.png

Intricate set: population / temporal (fig_pop_*)



- What: cycle-to-cycle timing regularity.
- How: inter-peak interval (IPI) vs cycle, a Poincare map (IPI n vs IPI n+1) with an SD1/SD2 ellipse, the IPI histogram, and a running mean.
- How to read: a tight cluster on the Poincare map and small SD1 = a very regular, low-jitter rhythm. SD1 is short-term variability, SD2 long-term.

fig_pop_patch_overview.png

Intricate set: population / temporal (fig_pop_*)



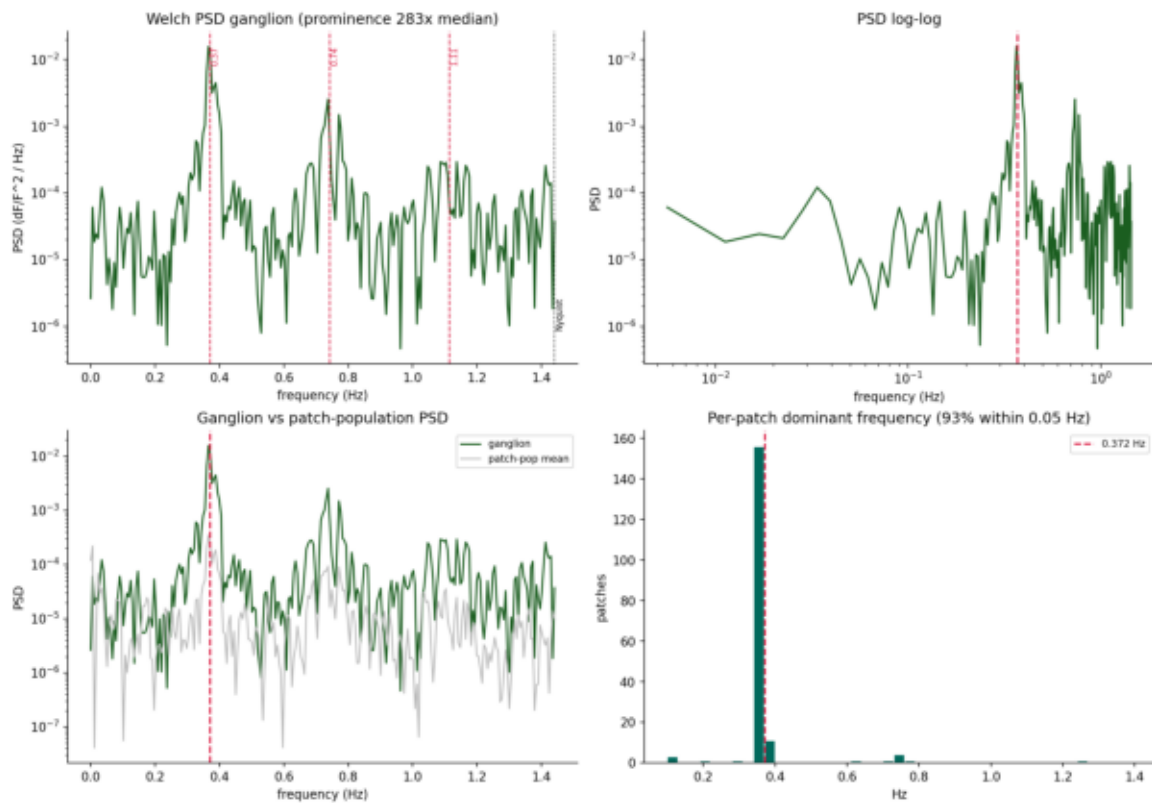
- What: small-multiples of the 36 most active patches.
- How to read: each mini-panel is one patch's dF/F on a shared scale. The rhythm
- looks nearly identical across panels, showing how uniform it is across the ganglion.

- ---

Intricate set: spectral (fig_spec_*)

fig_spec_psd_detailed.png

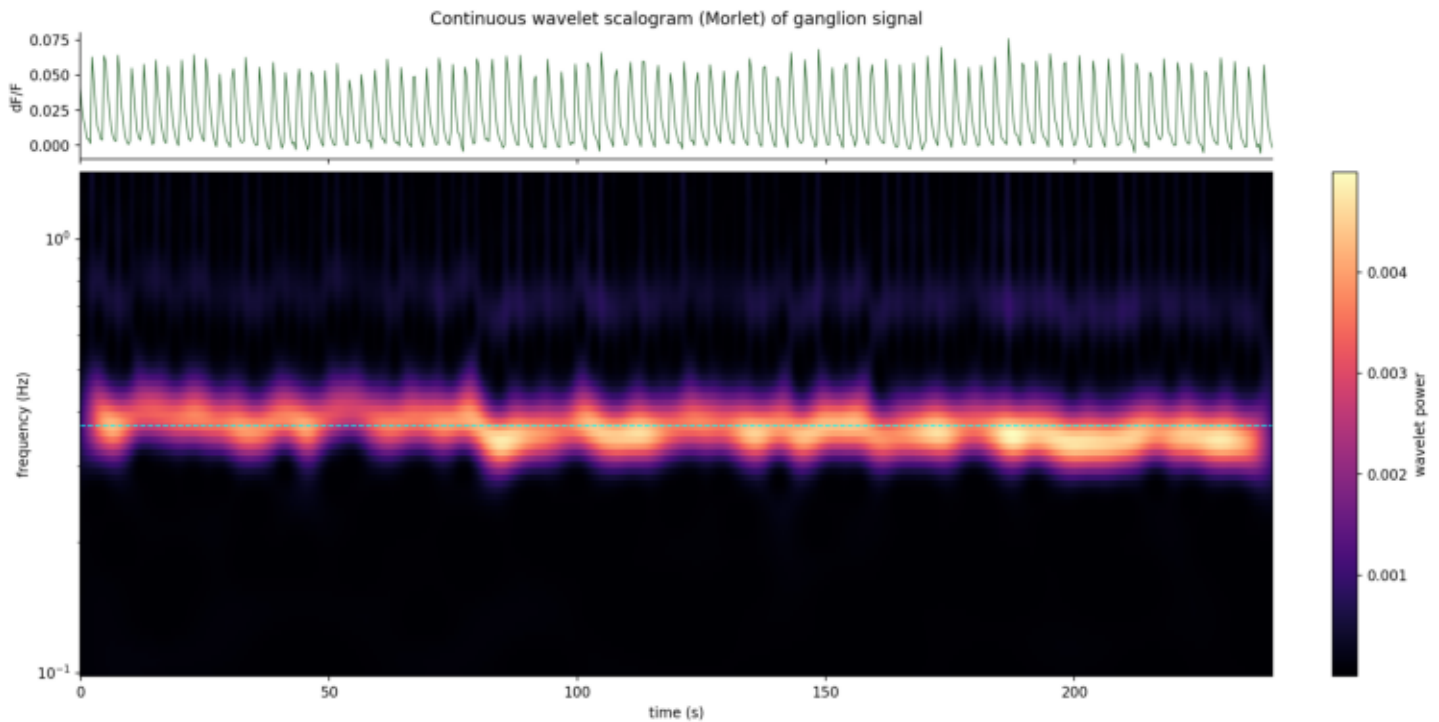
Intricate set: spectral (fig_spec_*)



- What: a detailed look at the power spectrum.
- How: Welch PSD on linear-log and log-log axes, with fundamental and 2nd/3rd harmonics marked; ganglion vs patch-population overlay; and a histogram of each of the
- 180 patches' own dominant frequency.
- How to read: the harmonic ladder confirms a periodic non-sinusoidal rhythm. The
- histogram shows 93% of patches peak within 0.05 Hz of 0.37 Hz, i.e. they share one
- frequency.

fig_spec_wavelet.png

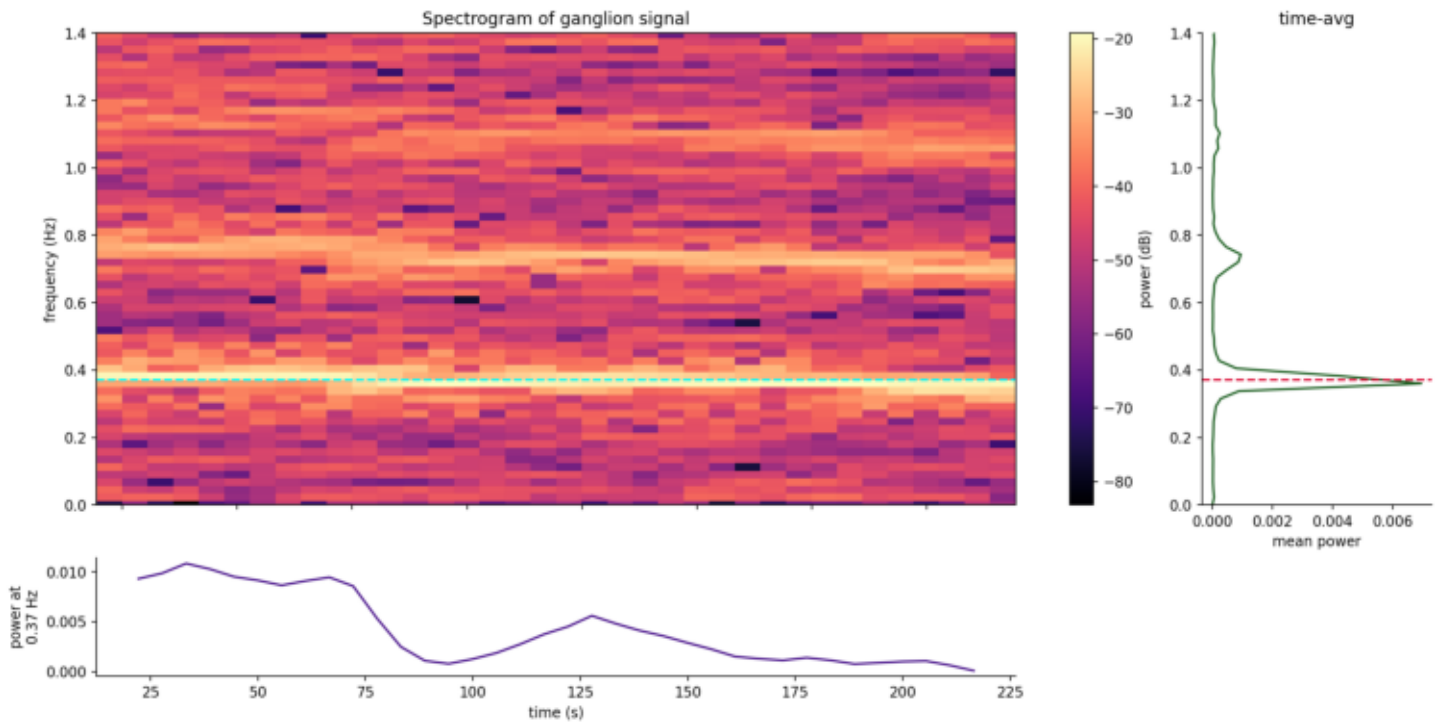
Intricate set: spectral (fig_spec_*)



- What: a time-frequency (wavelet) view of the ganglion signal.
- How: a Morlet continuous wavelet transform (implemented manually); the raw signal is aligned above it.
- How to read: a continuous bright horizontal stripe at 0.37 Hz for the full duration
- = the rhythm is present and stationary the entire time (complements the spectrogram with better low-frequency resolution).

fig_spec_spectrogram_detailed.png

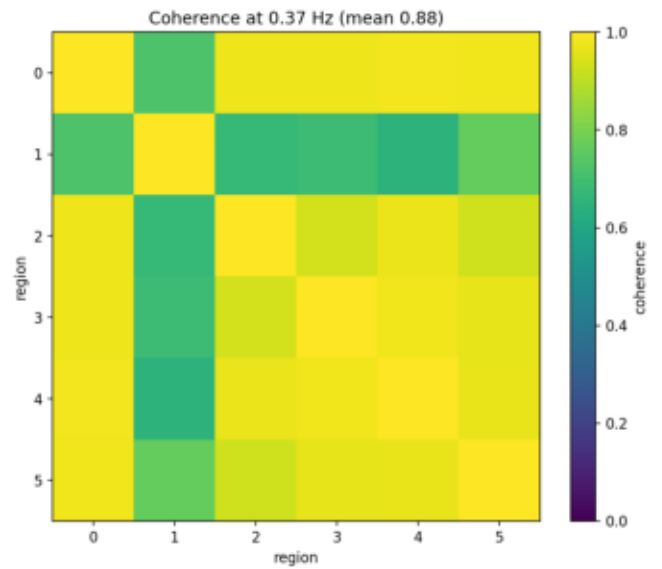
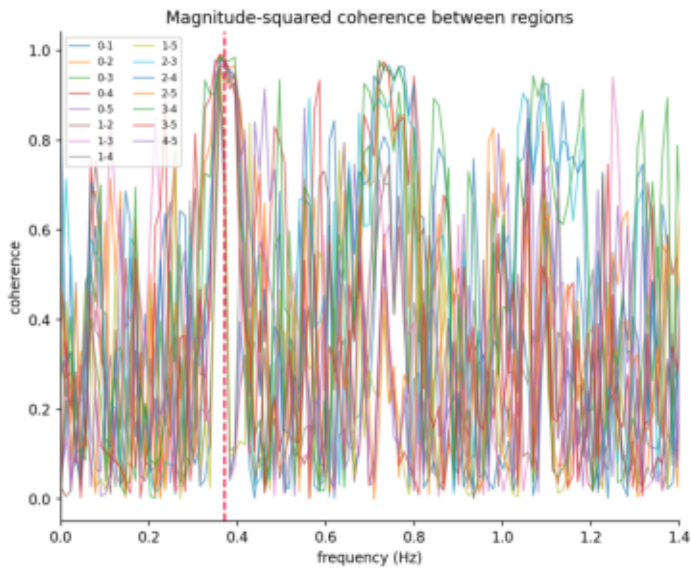
Intricate set: spectral (fig_spec_*)



- What: a high-resolution spectrogram with marginal summaries.
- How: scipy spectrogram with generous overlap; side panel = time-averaged spectrum,
- bottom panel = power at 0.37 Hz over time.
- How to read: stable horizontal band = fixed frequency; the bottom panel shows how
- the rhythm's strength fluctuates mildly over the recording.

fig_spec_coherence.png

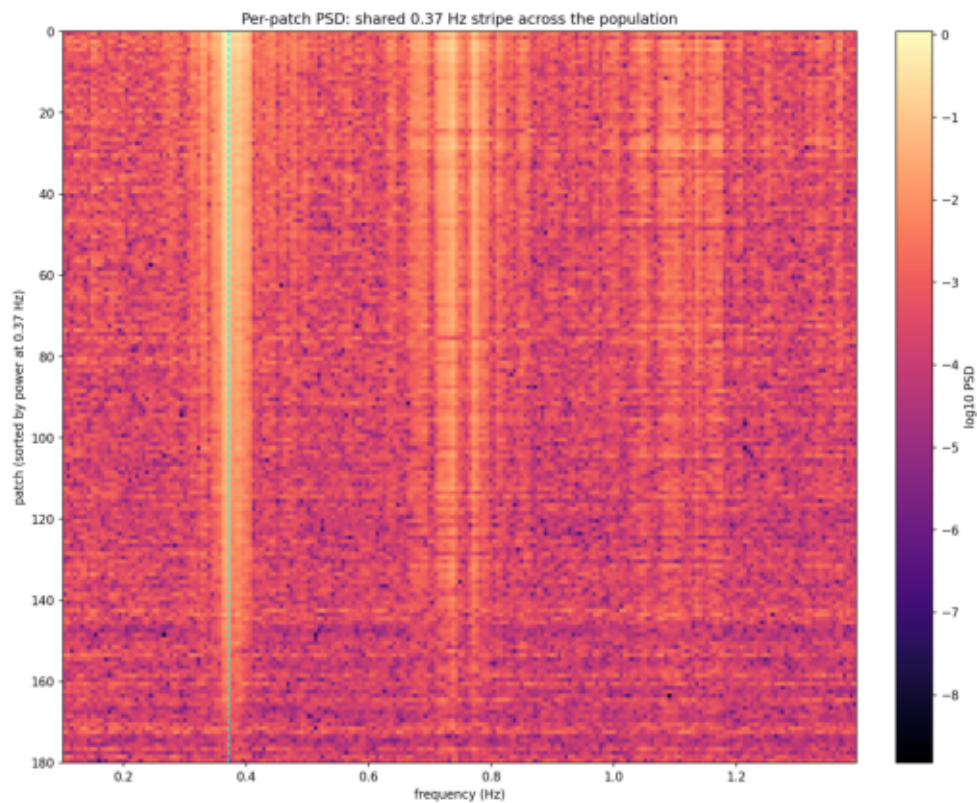
Intricate set: spectral (fig_spec_*)



- What: how strongly separated regions oscillate together, frequency by frequency.
- How: magnitude-squared coherence between 6 spatial region pairs, plus a matrix of coherence at 0.37 Hz.
- How to read: coherence near 1 at 0.37 Hz (mean ~ 0.88) means the regions are tightly phase-locked at the rhythm frequency, even though individual noisy patches are not.

fig_spec_patch_spectra.png

Intricate set: spectral (fig_spec_*)

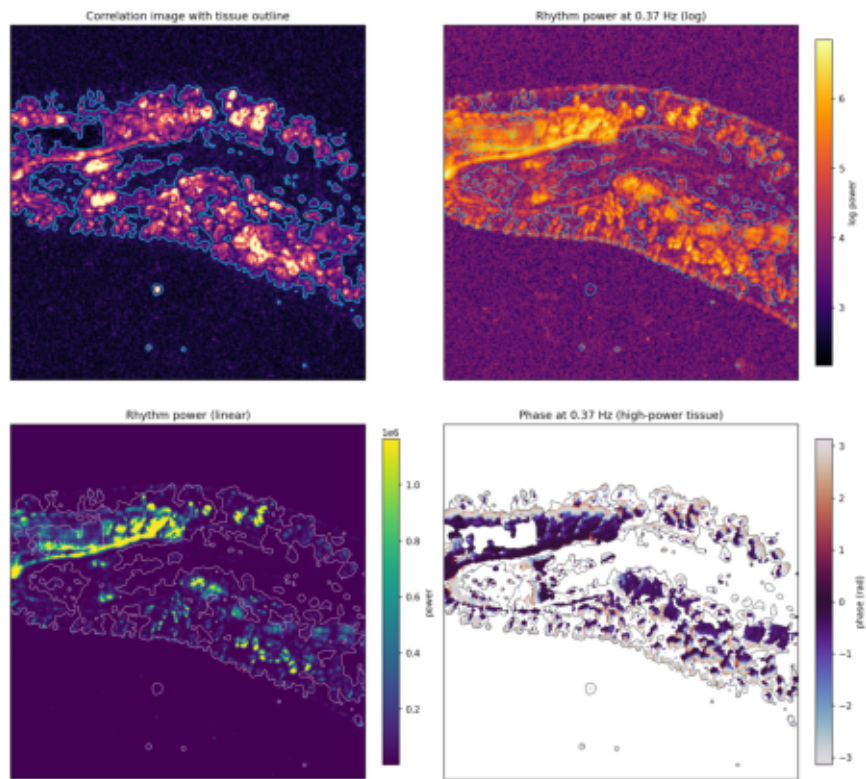


- What: every patch's spectrum stacked into one image.
- How: rows = 180 patches (sorted by power at 0.37 Hz), columns = frequency, color = log power; a vertical line marks 0.37 Hz.
- How to read: a sharp vertical stripe at 0.37 Hz shared down the rows = the whole population carries the same rhythm.
- ---

Intricate set: spatial (fig_spatial_*)

fig_spatial_power_phase.png

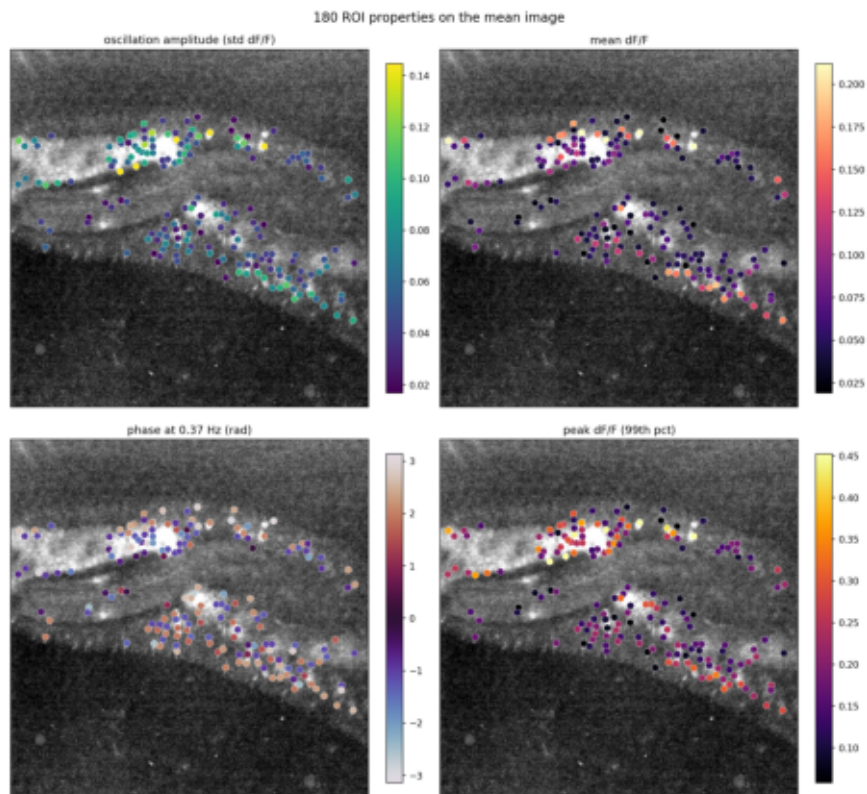
Intricate set: spatial (fig_spatial_*)



- What: a 2x2 of correlation image, rhythm power (log and linear), and phase.
- How to read: power panels show where the rhythm is strong; the phase panel (cyclic colors) is fairly uniform over the band = in-phase oscillation.

fig_spatial_roi_properties.png

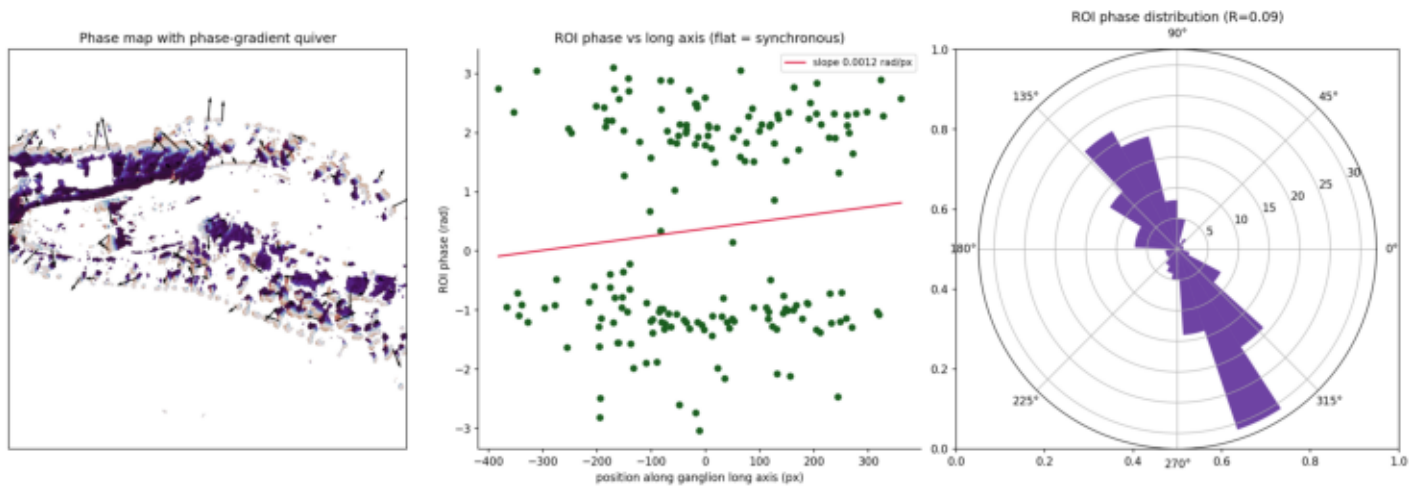
Intricate set: spatial (fig_spatial_*)



- What: the 180 ROIs colored by four properties on the anatomy.
- How: amplitude (std dF/F), mean dF/F, phase at 0.37 Hz, and peak dF/F per ROI.
- How to read: the amplitude/peak panels show which patches drive the rhythm; the
- phase panel shows whether timing varies in space (it is largely uniform among the
- strong patches).

fig_spatial_phase_gradient.png

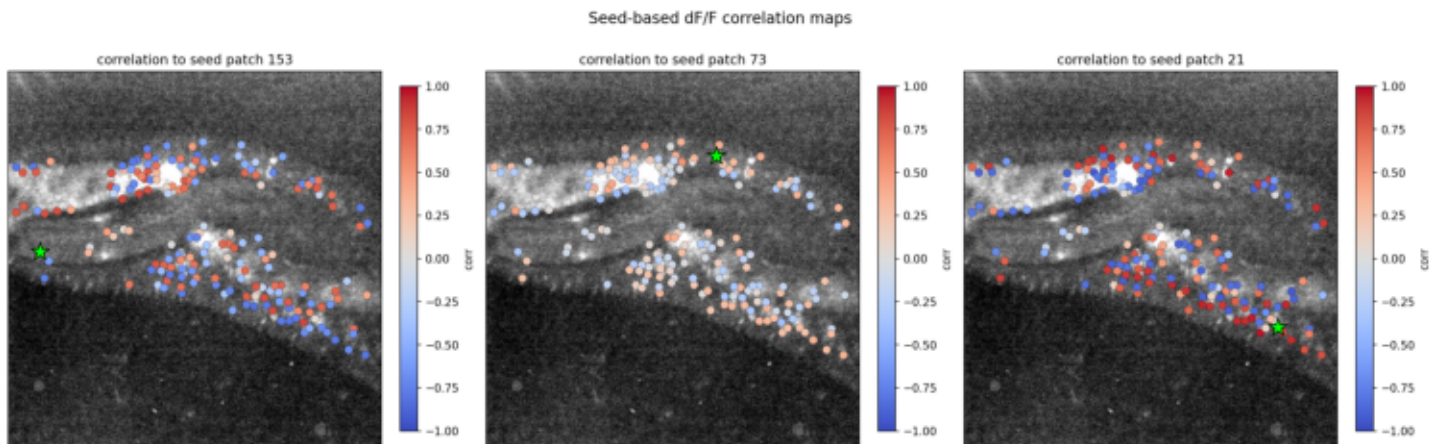
Intricate set: spatial (fig_spatial_*)



- What: an explicit traveling-wave test.
- How: phase map with a gradient quiver; ROI phase vs position along the ganglion long axis (PCA) with a regression slope; a polar histogram of ROI phases.
- How to read: a near-zero slope and no organized quiver direction = no traveling wave. Note the ROI phase histogram is fairly dispersed (resultant $R \sim 0.09$) because many weak patches have noisy phase; the strong patches and region averages are the coherent part.

fig_spatial_seed_correlation.png

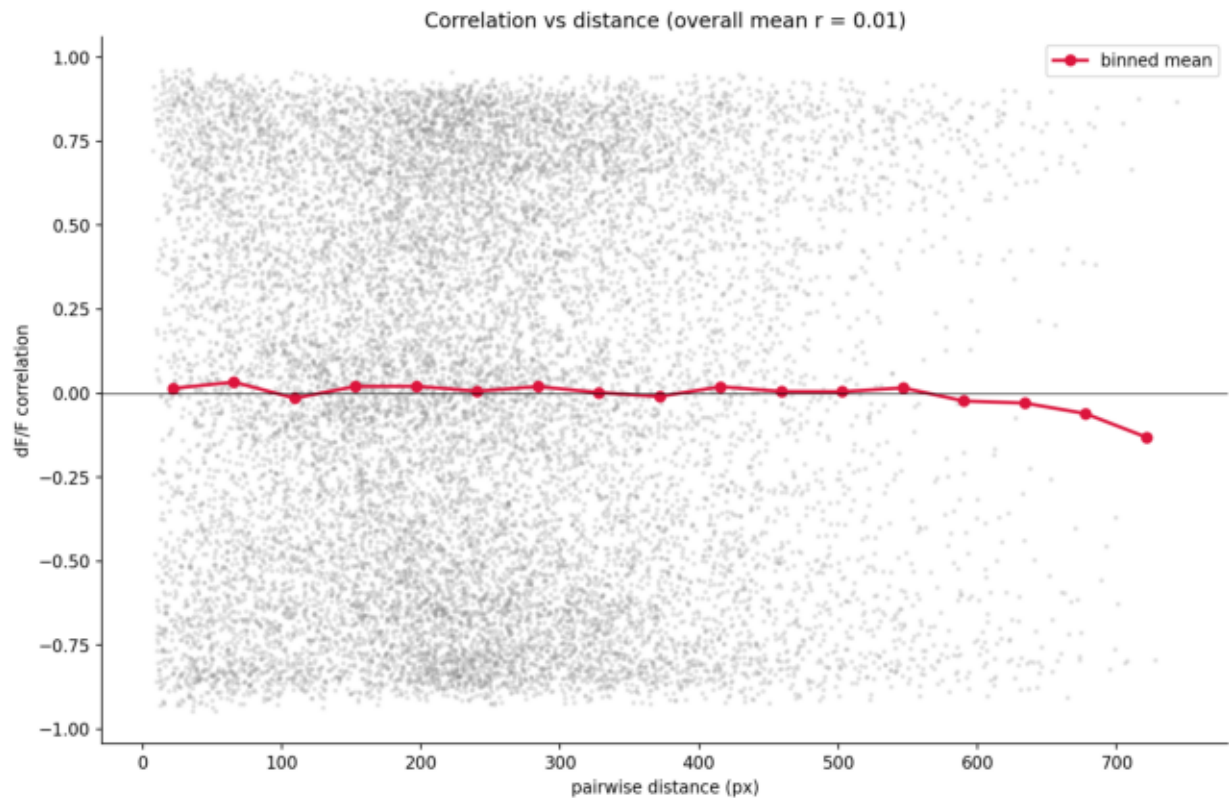
Intricate set: spatial (fig_spatial_*)



- What: how far co-activity reaches from a chosen patch.
- How: pick 3 seed ROIs in different regions; color all ROIs by their dF/F correlation to each seed.
- How to read: warm colors far from the seed = long-range coupling; here correlation is patchy rather than a smooth local halo, consistent with a global rhythm plus per-patch noise.

fig_spatial_distance_correlation.png

Intricate set: spatial (fig_spatial_*)

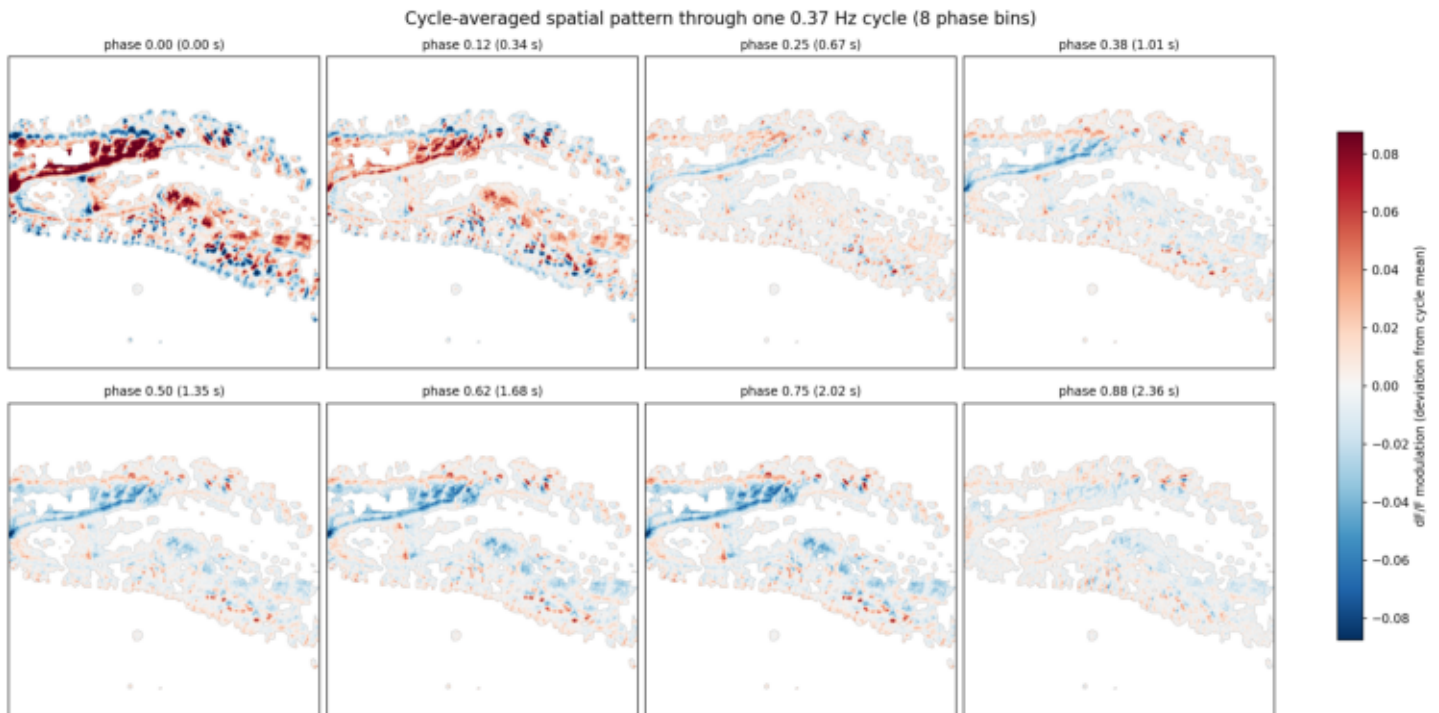


- What: does proximity predict co-activity.
- How: every patch-pair's correlation vs their centroid distance, with a binned-mean line.
- How to read: a flat binned line near zero = nearby patches are not much more correlated than distant ones, i.e. the coupling is global, not locally clustered (the overall pairwise mean is low because weak patches dominate the pair count).
- ---

**Intricate set: cycle-resolved spatiotemporal
(fig_cycle_*)**

fig_cycle_phase_montage.png

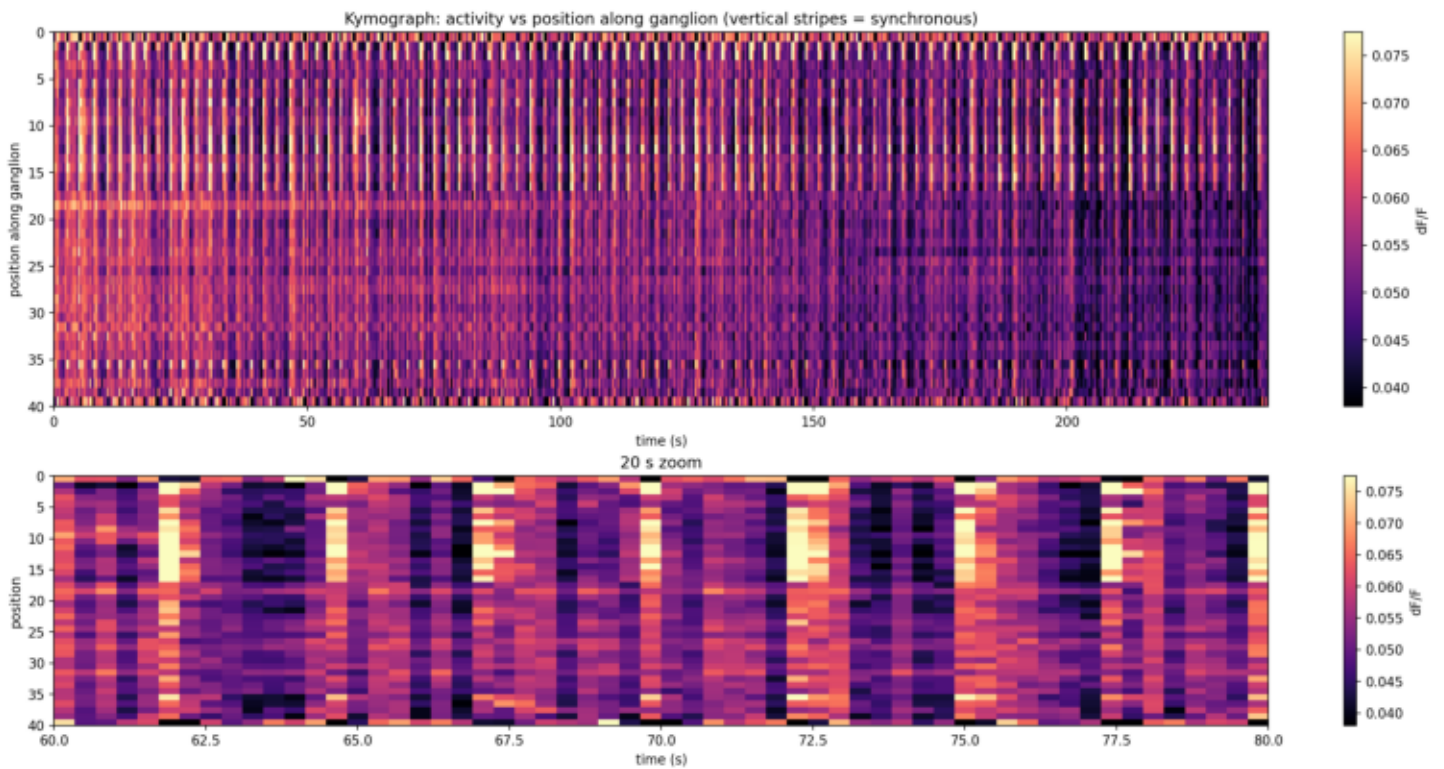
Intricate set: cycle-resolved spatiotemporal (fig_cycle_*)



- What: one average oscillation cycle played out in space.
- How: assign every frame a phase 0 to 1 within its cycle, bin into 8 phases, average the frames in each bin, and show the deviation from the cycle mean (red = above, blue = below).
- How to read: the whole ganglion turns red together near phase 0 and blue together near phase 0.5 to 0.75. Brightening/dimming as a unit = a standing, synchronous oscillation, not a wave sweeping across.

fig_cycle_kymograph.png

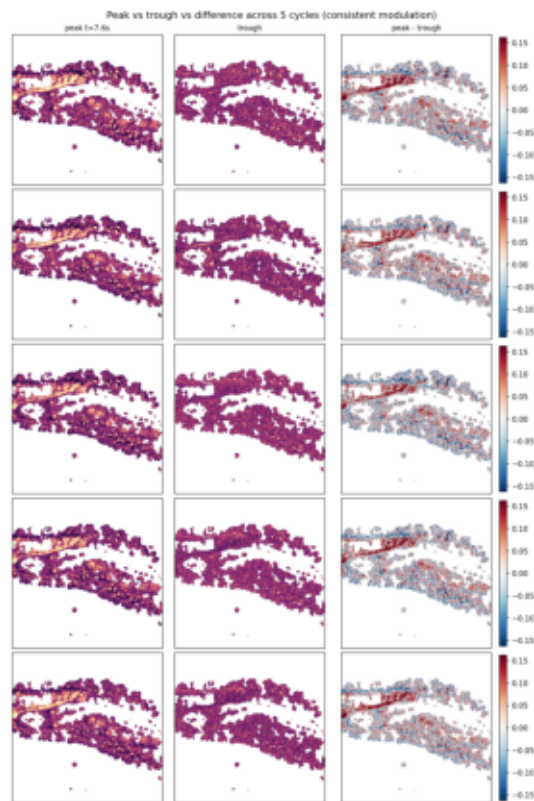
Intricate set: cycle-resolved spatiotemporal (fig_cycle_*)



- What: space (along the ganglion) vs time, the clearest wave test.
- How: project activity onto 40 bins along the ganglion long axis; x = time, y = position, color = activity. A 20 s zoom is below.
- How to read: vertical stripes = all positions peak at the same time (synchronous). Tilted stripes would mean a traveling wave. Here the stripes are vertical.

fig_cycle_peak_trough_gallery.png

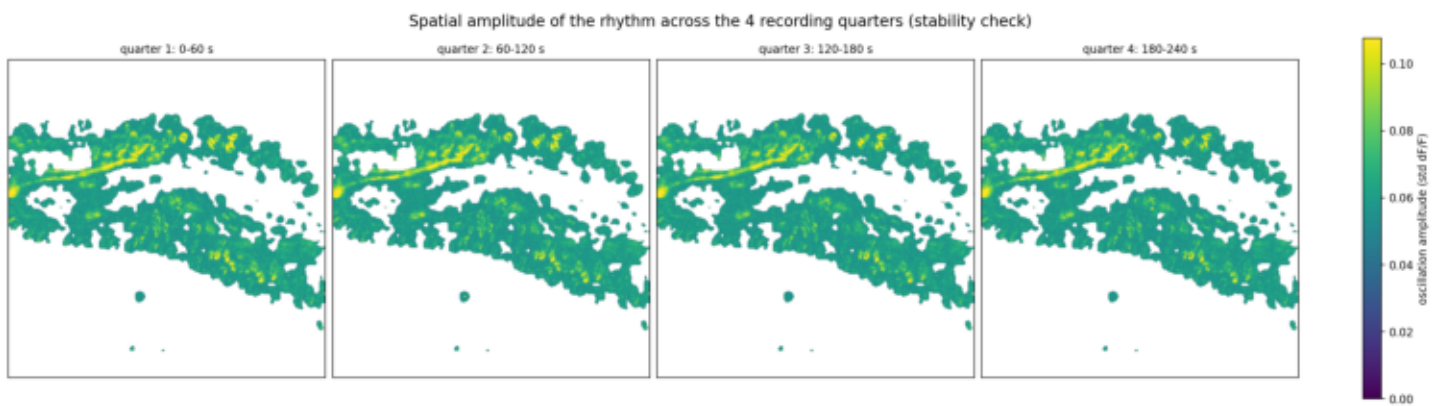
Intricate set: cycle-resolved spatiotemporal (fig_cycle_*)



- What: cycle-to-cycle consistency of the spatial pattern.
- How: for 5 separate cycles, show the peak frame, the trough frame, and their difference.
- How to read: peaks brighter than troughs, and the difference maps (right column)
- showing the same positive pattern across all 5 rows = the modulation repeats reliably.

fig_cycle_amplitude_map_over_time.png

Intricate set: cycle-resolved spatiotemporal (fig_cycle_*)



- What: is the rhythm's spatial amplitude stable over the recording.
- How: split into 4 equal time quarters; per-pixel oscillation amplitude (std over time) for each.
- How to read: similar maps across the four quarters = the rhythm neither grows nor fades nor moves over the 240 s.

fig_cycle_movie.mp4

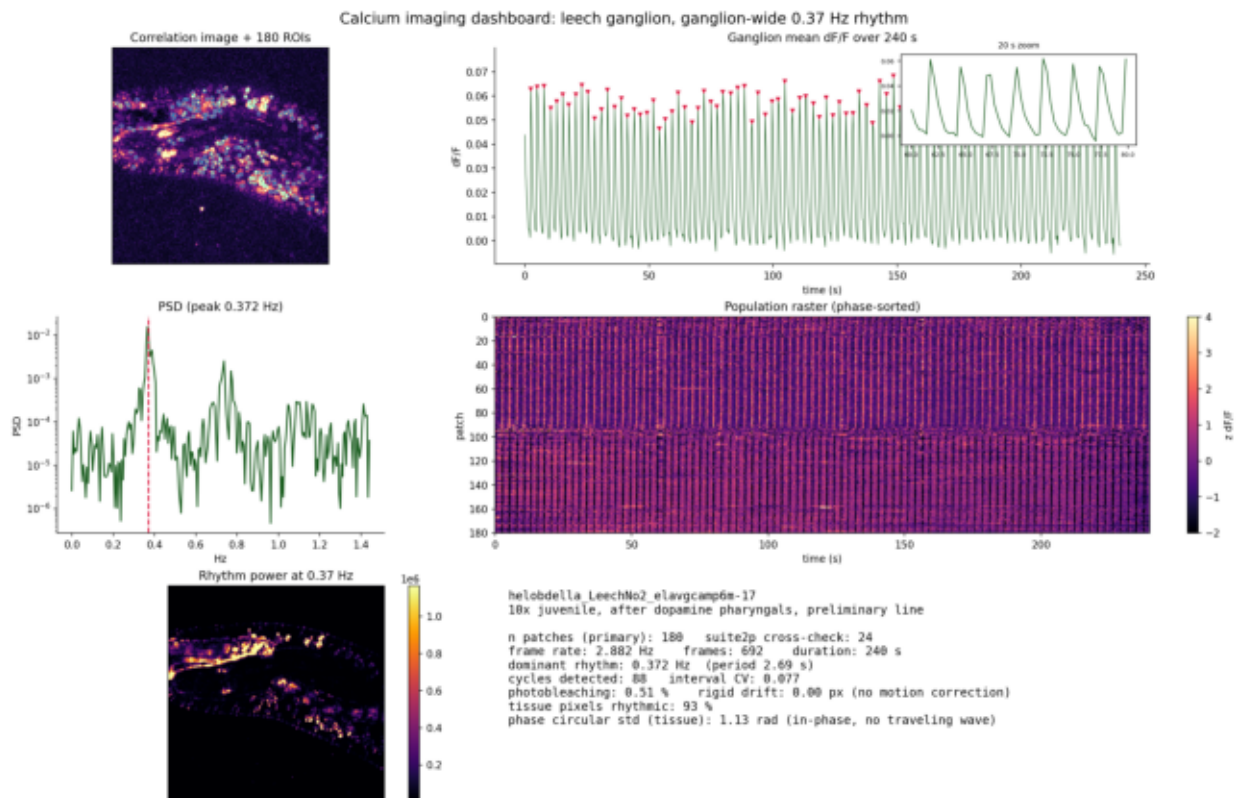
Intricate set: cycle-resolved spatiotemporal (fig_cycle_*)

- What: the cycle-averaged sequence (the 8 phase bins above) looped a few times as a
- short movie.
- How to read: watch the ganglion pulse brighter and dimmer as a single unit; this is
- the rhythm in motion. (Open the mp4 directly; it is not an inline image.)
- ---

Intricate set: dashboards and QC (fig_dash_*)

fig_dash_master.png

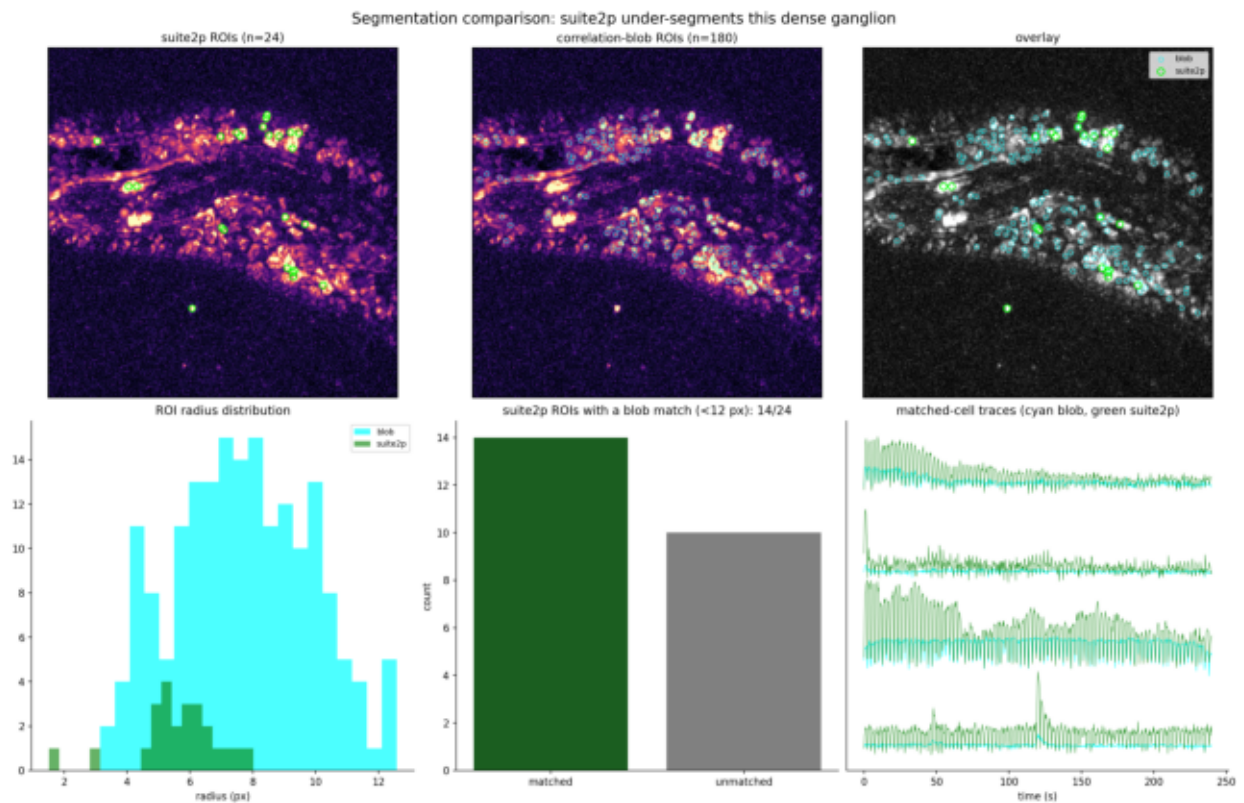
Intricate set: dashboards and QC (fig_dash_*)



- What: the one-page summary of the whole analysis.
- How: combines the ROI map, ganglion trace with cycle peaks, PSD, phase-sorted
- population raster, rhythm power map, and a text block of key metrics.
- How to read: start here for the overview, then drill into the themed figures. The
- text panel lists n patches, frequency/period, CV, bleaching, drift, and phase spread.

fig_dash_segmentation_compare.png

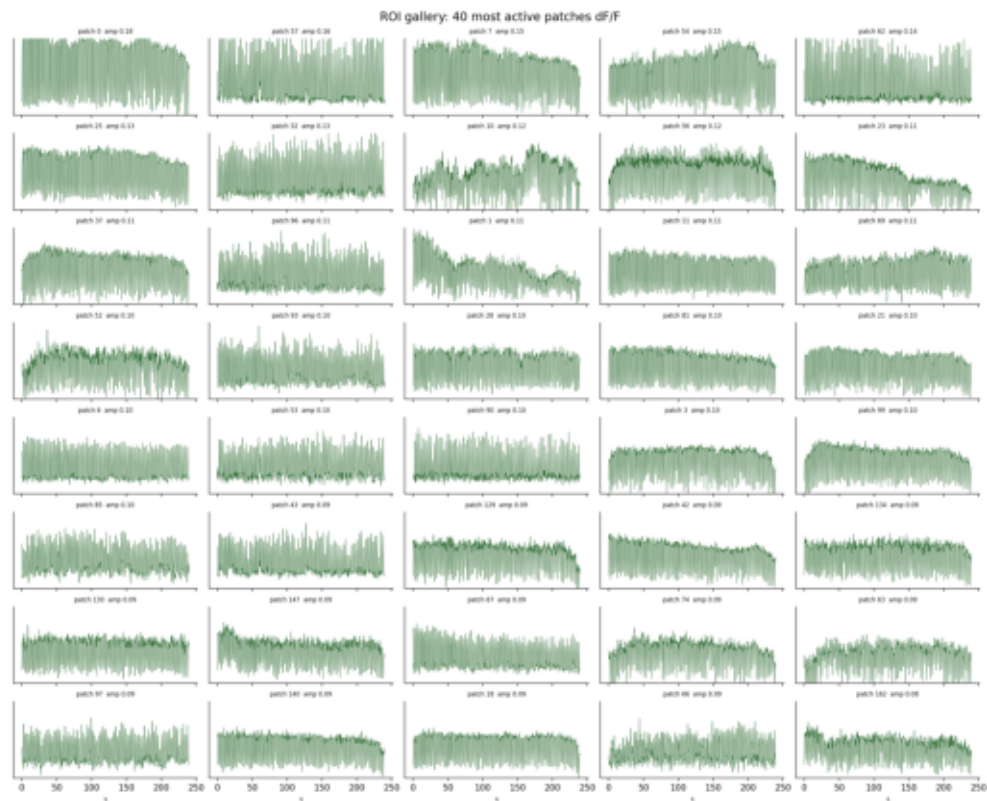
Intricate set: dashboards and QC (fig_dash_*)



- What: a detailed suite2p vs correlation-blob comparison.
- How: both ROI sets on the correlation image and overlaid; ROI size histograms; how
- many suite2p ROIs have a nearby blob match; and example traces from cells found by both.
- How to read: the overlay and the matched count (14 of 24 suite2p ROIs have a blob
- within 12 px) show the methods agree where they overlap, but the blob method captures
- far more of the visible cells.

fig_dash_roi_gallery.png

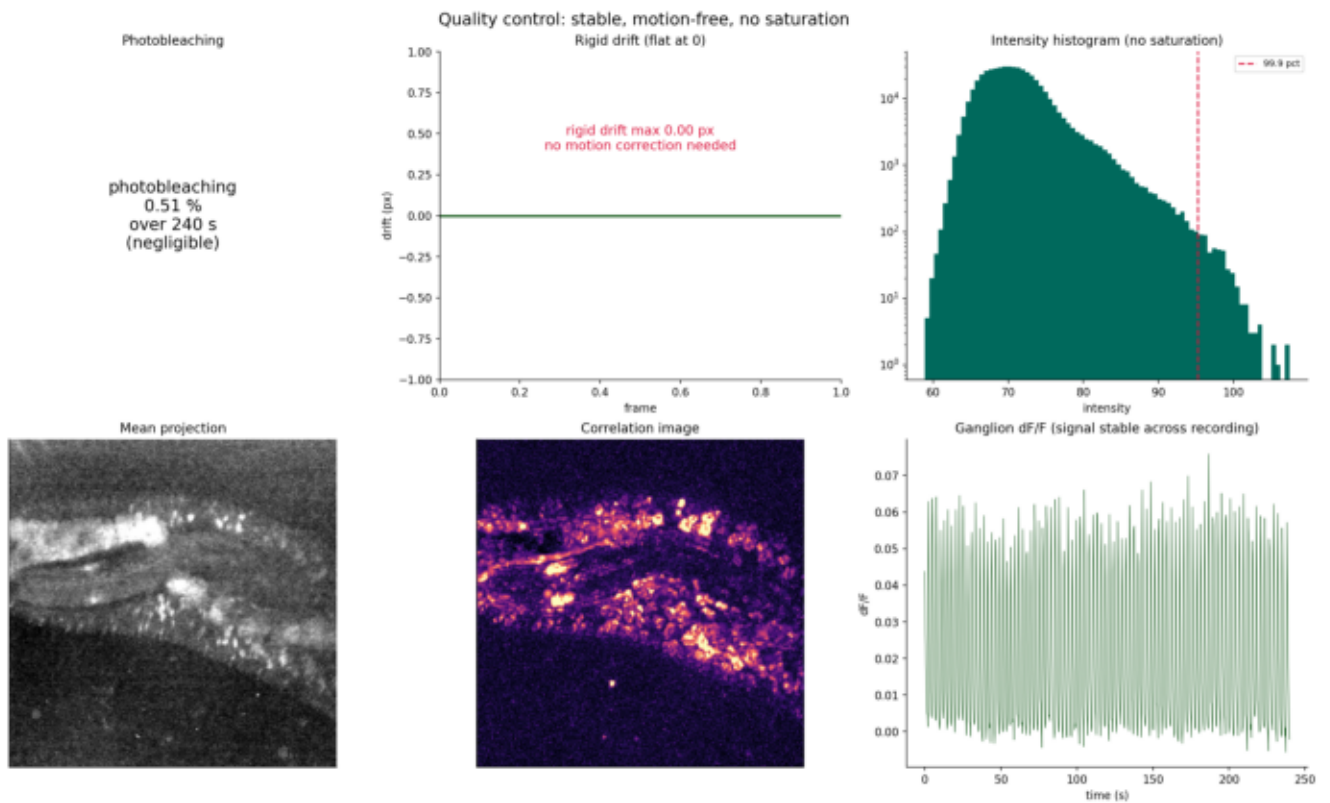
Intricate set: dashboards and QC (fig_dash_*)



- What: a contact sheet of the 40 most active patches.
- How to read: each tile is one patch's dF/F (shared scale) with its id and amplitude;
- a quick way to eyeball signal quality and confirm the rhythm is widespread.

fig_dash_qc_panel.png

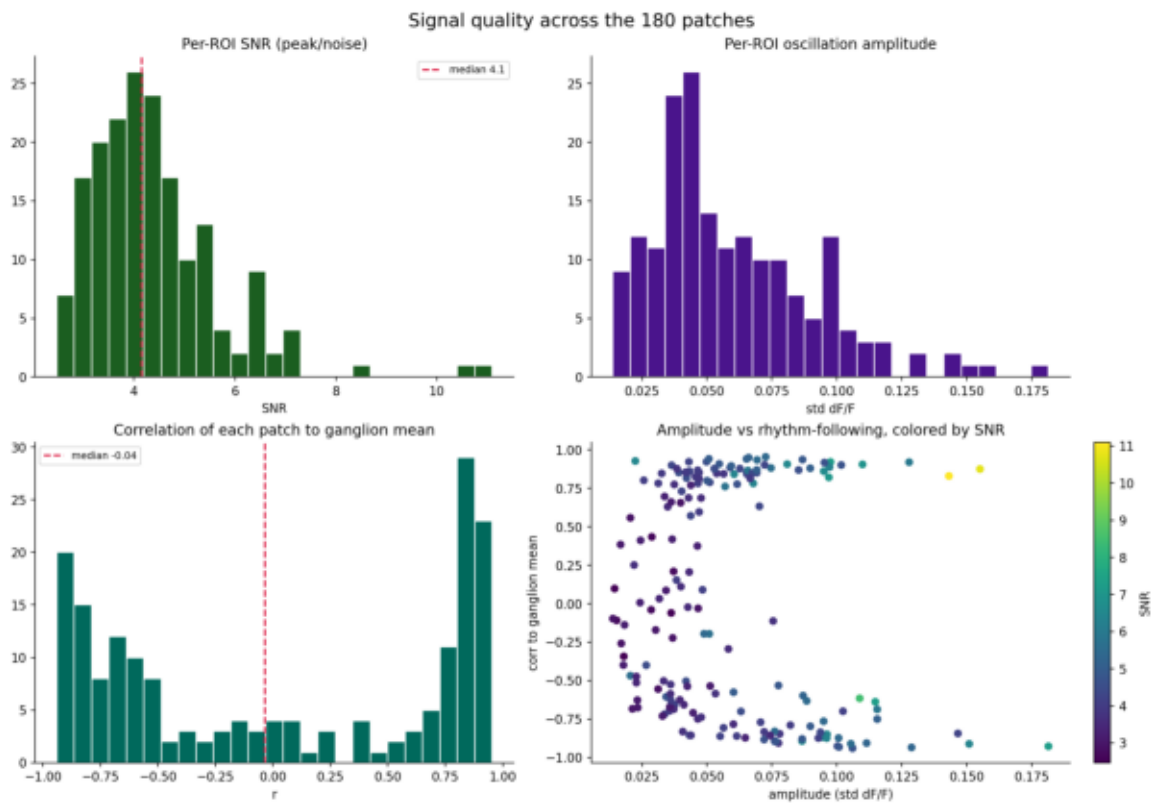
Intricate set: dashboards and QC (fig_dash_*)



- What: a polished recap of the quality control.
- How to read: bleaching negligible, drift flat at 0 (no motion correction), intensity histogram with no saturation, and the mean/correlation projections. Confirms the data are clean to analyze directly.

fig_dash_trace_quality.png

Intricate set: dashboards and QC (fig_dash_*)



- What: how good the per-patch signals are.
- How: distributions of per-ROI SNR, oscillation amplitude, and correlation to the ganglion mean; plus amplitude vs rhythm-following colored by SNR.
- How to read: median SNR ~ 4.1 ; about 40% of patches follow the ganglion mean at $r > 0.5$. This is the key nuance: a strong rhythmic subset drives the clean ganglion signal, while many weaker patches are noisy (expected at 10x, low photon counts).
- ---

Reading order suggestion

1. fig_dash_master.png for the overview.
2. s1_motion.png + fig_dash_qc_panel.png to trust the data (no motion, no bleaching).
3. s2b_method_comparison.png to understand the ROIs.
4. s3_spectrum.png + fig_spec_wavelet.png for the rhythm frequency and its stability.
5. fig_cycle_kymograph.png + fig_cycle_phase_montage.png for the synchronous, standing nature of the oscillation.
6. fig_dash_trace_quality.png for the caveat that a rhythmic subset carries the signal.

Caveats that apply to every figure

- 10x, preliminary line: interpret at the ganglion/regional level, not single neurons.
- "After dopamine pharyngals" but no pre-dopamine baseline exists here, so no causal claim about dopamine driving the rhythm.
- Sampling at 2.882 Hz resolves the 0.37 Hz rhythm well (~ 8 samples/cycle) but anything above ~ 1.4 Hz would be aliased.